

**A DEEP LEARNING-ENHANCED STATISTICAL MACHINE
TRANSLATION OF FILIPINO LOCAL LANGUAGES**



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A Deep Learning-Enhanced Statistical Machine Translation of Filipino Local Languages

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Abstract

Within the Natural Language Processing (NLP) field, this study investigated the potential enhancement of Statistical Machine Translation by Bidirectional Long Short-Term Memory (Bi-LSTM). The linguistic diversity in the Philippines posed significant language barriers, particularly in rural areas where standardized Filipino is less prevalent. Data was gathered from dictionaries, children's stories, rhyme books, and ChatGPT 4 prompts. The researchers employed a statistical method using Python as its medium, assigning probabilities to words from the source to the target language. Results showed improved accuracy and higher probability scores, as evidenced by increased BLEU (Bilingual Evaluation Understudy) metrics. These findings indicated that incorporating Bi-LSTM into Statistical Machine Translation systems enhances translation quality, suggesting a promising approach for overcoming linguistic challenges in multilingual settings.

Categories and Subject Descriptors

[Deep Learning]: Subset of Machine Learning - Artificial Neural Network

General Terms

Languages, Algorithms, Documentation, Performance.

Keywords

Machine Translation; Statistical Machine Translation; Bi-LSTM; LSTM; Languages.

1. INTRODUCTION

1.1 Background of the Study

The Philippines is an island country in Southeast Asia, and its culture and history are diverse. There are many spoken languages and dialects, such as Hiligaynon and Tagalog, due to the diversity of its people. Hiligaynon is the language spoken in the Western Visayas region, and it is also known as Ilonggo to the locals. The province of Iloilo, Negros Occidental, Capiz, and Guimaras has the largest population of native speakers, and it is the second most widely spoken language in the country. Filipino is the standardized form of Tagalog and one of the major

languages spoken in the Philippines: the national language and the lingua franca unite the nation's diverse linguistic and cultural communities. People primarily speak Filipino in the Luzon region.

Hiligaynon and Filipino are the two languages that are rich in cultural and linguistic subtleties, which presents a unique and complex pair for machine translation. The two languages are distinct yet similar due to shared cultural and historical influences since they came from the same language family, the Austronesian Family of Language. The problem with achieving a correct translation is that machines need help picking up on the nuances of speech. Something spoken in Hiligaynon translated into Tagalog or Filipino will sometimes be broken in either grammar, structure, or both. The complexity of these languages necessitates a translation system using Statistical Machine Translation to capture not only the semantic meaning but also the cultural context within expressions of the sentences.

One popular model well-known for its capacity to incorporate context from previous and subsequent items in a data sequence is the Bidirectional Long Short-Term Memory (Bi-LSTM) model. This distinct quality has excellent potential for overcoming the difficulties in translating between Filipino and Hiligaynon. The Bi-LSTM's bidirectionality enables the model to fully comprehend the linguistic environment by deftly grasping the nuances of the language's syntax and semantics. Integrating Bi-LSTM into a statistical machine translation system represents a significant advancement. The translation model can consider both the present input and the language elements that will be added in the future with this addition. This forward and backward information flow makes a more sophisticated understanding of the original language possible, improving the system's capacity to provide linguistically and culturally accurate translations.

Given that the Bi-LSTM is anticipatory, the translation model can anticipate and capture Hiligaynon and Filipino-specific idioms, cultural allusions, and subtleties. As a result, the translated result is expected to be more dependable, and its message will align with the cultural quirks in the original text. Incorporating Bi-LSTM into the translation framework is a powerful instrument for improving the standard of cross-linguistic communication, cultivating linguistic precision, and encouraging a more comprehensive comprehension of the cultural background in machine translation.

This study aims to improve the accuracy of SMT or statistical machine translation using bidirectional-LSTM with added layers tailored for Hiligaynon and Filipino. The researchers will use

Hiligaynon and Filipino corpus to train and evaluate Bi-LSTM performance. The expected outcome of the study is the potential improvements in the translation quality, mainly the accuracy of the translation system. It also brings advancement in technology to achieve and promote cross-cultural communication. By adding Bidirectional Long Short-Term Memory (Bi-LSTM) models with specialized extra layers created especially for the Hiligaynon and Filipino languages, the study aims to improve the accuracy of Statistical Machine Translation (SMT). The main goal is to use Hiligaynon and Filipino corpora to train and evaluate the Bi-LSTM model's performance in machine translation. Additionally, including specialized layers is anticipated to improve the model's flexibility in response to Filipino and Hiligaynon's distinct syntactic and semantic structures, guaranteeing a translation of translated information more suited for its context. Ultimately, the project hopes to further cultural exchange and efficient communication across many language communities and significantly contribute to machine translation.

For a considerable amount of time now, machine translation has been utilized to translate different languages. Long-and Short-Term Memory, or LSTM, is one of these machine translation technologies. Numerous researchers have undertaken to improve its application in the domains in which many projects currently employ this method. Over the past ten years, there has been a hunt for better LSTMs and, consequently, Bidirectional LSTMs (BLSMTs). These efforts often depend on giving the SMT data to train it and make it behave according to the instructions. The technique known as Bidirectional Long-Term Memory (BiLSTM) allows any neural network to process data from the past to the present and backward from the future to the past. [12]. In one instance, a Bidirectional LSTM is trained to assess code by examining several source codes, and it attains a 32% increase in accuracy rating. [13]. According to prior testing by related studies, bidirectional LSTM models do not work well as transformers inside recurrent networks. [15]. The researchers aim to reduce the number of issues the neural network encounters without altering its typical operation, such as resource requirements, training time lags, and errors, to increase the efficiency of the Bidirectional LSTM model. The project aims to improve research quality and accuracy for LSTM and related neural network advocates in the future.

It helps the advocates figure out a new approach to using the additional information from earlier research that attempted to accomplish this feat in conjunction with the neural network. Bidirectional LSTMs considerably impacted Part-Of-Speech (POS), which defines a group of words with comparable grammar, such as verbs and adjectives, and Named Entity Recognition (NER), which detects entities like nouns. According to the study, utilizing all three translated Vietnamese and Korean is more seamless. Regretfully, the study chose to determine the Translation Error Rate (TER) rather than explicitly search for accuracy, which makes it more difficult to compare the results with those of the other studies. What does stand out, though, is that TER significantly decreased after the introduction of a UTagger, which limits the ambiguity of certain words by coding in different tags for each meaning (ex. word_01, word_02), and that Bidirectional LSTMs scored exceptionally high accuracy during one of their training sets for just Vietnamese. The lower the TER, the more accurate the translation, but it also considers other factors that do not count as actual "accuracy." The study also suggests that future research can significantly boost accuracy by utilizing syntactic relationships. [13].

Compared to other deep-learning models, Bi-LSTM is significantly better at extracting context and meaning from

phrases. By including specific enrichments from layers and morphological characteristics, Bi-LSTMs can accurately locate what they seek in highly lexical data while overcoming obstacles that most deep learning models encounter, including unclear text arrangements. [5]. When evaluating various deep learning models for their ability to provide acceptable results, research that a configuration of the Neural Machine translation model has increased its accuracy via an accuracy score method called BLEU score; the study did use the Bi-LSTM model. The study discovered a light difference between the baseline NMT and Bi-LSTM. Models combining word and character achieved a higher BLEU score [9]. According to the study, their suggested model performed significantly better after receiving additional training and data. This brings us to the study model's weakness. If the necessary conditions are met, it will produce noticeably better results than the other baselines but will require a lot more data and training than those baselines. [7].

Alla Lo et al. used LSTM to translate from French to Senegalese in a case study. Researchers discovered that the case study matched their goals for their research well, which involved translating from Hiligaynon to Filipino and vice versa using LSTM. In terms of accuracy, the study discovered a few crucial characteristics of LSTMs and the layers they employed. The least accurate model was the baseline LSTM, with a mere 56.69%. A noteworthy 63.89% increase in the LSTM's accuracy by adding an attention layer. With a baseline accuracy score of 68.03%, this case study identifies bidirectionality as an additional layer to the LSTM, a significant improvement over LSTM + attention. Finally, they attempted it with bidirectional, extra layers, and attention, and the accuracy score was 72.27%. [1]. The results demonstrate that a Bidirectional LSTM can outperform a baseline with the same amount of training in terms of accuracy even when it does not have the highest accuracy statistics when more layers are added.

1.2 Purpose and Description

Integrating Bi-LSTM into a statistical machine translation model allows the researchers to analyze and test its performance and provide a result that will contribute to the body of knowledge in computer science.

This study will use NLP, or Natural Language Processing, to further technological advancements. The researchers will use and optimize the current bidirectional LSTM design. Researchers will also use statistical machine translation to identify and classify Filipino and Hiligaynon terms precisely in the Filipino language.

1.2.1 General Objectives

The Primary objective of this research is to develop a machine translation system focused on translating Hiligaynon to Tagalog using a Bidirectional Long Short-Term Memory (Bi-LSTM) network. By Implementing a layered-Bi-LSTM architecture, the model is expected to improve handling word-for-word accuracy in Hiligaynon to Filipino and vice versa.

This study aims to enhance translation accuracy between two languages using Statistical Machine Translation with a Bi-LSTM model with multiple layers. Furthermore, Bi-LSTM is designed to handle sequential dependencies and bidirectional context better, resulting in more accurate translation with layers.

1.2.2 Specific Objectives

1.2.2.1 To create a model for statistical machine translation that uses a bidirectional LSTM to change to Hiligaynon and Filipino only.

1.2.2.2 To tokenize and normalize text to carry out data pre-processing and clean the necessary resources that call for an effective word translation between the Hiligaynon and Filipino languages from dictionaries and open-source data sets.

1.2.2.3 Using the data sets provided, run a simulation using the baseline and suggest a Bidirectional LSTM Statistical Machine Translation model.

1.2.2.4 The objective is to evaluate the experimental findings between the baseline and suggested models to determine which model performs better in accuracy when translating text in Filipino and Hiligaynon.

1.3 Scope and Limitation

1.3.1 Researchers aim to advance the effectiveness of named entity recognition (NER) by leveraging statistical machine translation, with a specific emphasis on optimizing bidirectional Long-Short-Term Memory networks (LSTMs). The study primarily draws on linguistic data from Hiligaynon and Filipino dictionaries, with additional words and sentences from children's songs, stories, and prompts generated by Chat GPT-4. Native speakers meticulously validate all data to ensure accuracy and cultural relevance.

1.3.2 This study is limited to the integration of statistical machine translation techniques within the framework of bidirectional LSTMs, focusing exclusively on the Filipino and Hiligaynon languages. This experimental research does not extend beyond those specified to other languages, dialects, or machine learning models. Additionally, broader topics unrelated to the core objective of enhancing named entity recognition (NER) through bidirectional LSTMs are outside the purview of this study.

2. METHODS

This study will conduct experimental research on a suggested model to improve the Bidirectional LSTM's capacity to translate text between Filipino and Hiligaynon vernacular using statistical and mathematical translation. Using a similar method, the researchers will compare the study's improvement with the baseline algorithm.

2.1 Data Collection Method

The researchers intend to gather information employing dictionaries, open-source data collections, and Filipino textbooks. Furthermore, OpenSubtitle, one of the current data sets, covers over 60 languages, including Filipino and other languages. A set of parallel multilingual corpora is called OpenSubtitle. The dataset comprises 1689 bitexts covering 2.6 billion phrases in 60 languages and gathered from an extensive database of TV and movie subtitles.

2.1.1 Datasets for the Study

Filipino/Tagalog Corpuz

The datasets consist of Filipino/Tagalog words and sentences with their correct definitions and grammar. Words were taken from a Filipino Dictionary available online [29].

Hiligaynon Corpuz

Datasets contain various words and sentences in Hiligaynon with correct definitions and grammar.

Additional words and sentences are created using Chat GPT 4 prompts validated and approved by a native speaker and checked with a Hiligaynon Dictionary [28]. Furthermore, the researchers also used various Filipino children's songs, stories, and rhymes.

2.2 Baseline Bidirectional LSTM

The Bi-LSTM model is mainly used in machine learning based on NLP or natural language processing, such as translation, sentence classification, and entity recognition. It comes from its predecessor, the LSTM or long- and short-term memory, which improves it by providing forward and backward directions for information instead of the conventional forward layer. Both directions are connected to the same output layer [9]. Furthermore, its architecture processes forward and backward information, which can be applied to many other sequence-based tasks in different branches of AI [11].

Furthermore, Bi-LSTM will run into input in forward LSTM and backward LSTM. The forward direction will process the data in the front, while the backward direction will process the data starting at the end. It will proceed when the two data are combined, and the information for forward and backward LSTM is preserved [11]. This model depends on both forward and backward information since, behind its elements, it can identify more than its counterpart, the conventional LSTM.

Below is the Mathematical equation for the process of Bi-LSTM.

Forward Direction (LSTM)

$$\begin{aligned} i_f &= \sigma(W_i * x + U_i * h_f + b_i) \\ f_f &= \sigma(W_f * x + U_f * h_f + b_f) \\ o_f &= \sigma(W_o * x + U_o * h_f + b_o) \\ c_{hat_f} &= \tanh(W_c * x + U_c * h_f + b_c) \\ c_f &= f_f * c_f + i_f * c_{hat_f} \end{aligned}$$

Expected Output:

$$h_f = o_f * \tanh(c_f)$$

Backward Direction (LSTM)

$$\begin{aligned} i_r &= \sigma(W_i * x + U_i * h_r + b_i) \\ f_r &= \sigma(W_f * x + U_f * h_r + b_f) \\ o_r &= \sigma(W_o * x + U_o * h_r + b_o) \\ c_{hat_r} &= \tanh(W_c * x + U_c * h_r + b_c) \\ c_r &= f_r * c_r + i_r * c_{hat_r} \end{aligned}$$

Expected Output:

$$h_r = o_r * \tanh(c_r)$$

Expected Final Output

$$h = (h_f; h_r)$$

The Bidirectional LSTM consists of forward and backward LSTM. The forward LSTM processes the input sequences in a normal direction as a conventional LSTM, while the backward LSTM processes the input in reverse. Both hidden states of LSTM concatenate to produce a final output that enables them

to capture information from both past and future sequences [5]. The process can be visualized in Figure 1 of the appendix page.

2.3 Bidirectional Statistical Translation

The researchers study an improved Bidirectional LSTM that shows on Figure 2 of the appendix page. It consists of four parts, which are: the Dataset, Data Cleaning and Pre-processing, Feature Selection, Extraction and Categorization and lastly, the Result. Each part has its own part in processing and handling the data. Data will be fed on the Machine Learning model. All of it except the Dataset will be explained throughout the entire section since it has already been described in Chapter 1.

2.3.1 Data Cleaning and Pre-processing

In this section, the sentence will be altered and turned into a proper form. The researchers focused on improving the model's accuracy, which identified the proper pre-processing method. Tokenization, Punctuation Removal, Stemming, and Word Normalization are known to provide possible solutions.

2.3.1.1 Tokenization

One of the well-known methods of breaking down words in Natural Language Processing is Tokenization. This is the first step when working with NLP, as it fragments each word or text into bits called a token [14]. The core purpose of Tokenization is to simplify each text and convert it into a format that is suitable for future processes. Most tokens will have meaning tied to it and will contribute to the overall meaning of a sentence. This will guide in pinpointing the sentence's meaning if integrated with other tokens [21].

2.3.1.2 Word Normalization

After Tokenization, word normalization will be used. It transforms words into a single consistent form to ensure different words are consistent and make it easier for the algorithm. It decreases the amount of information and enhances the efficiency of each word by categorizing individual tokens. During its process, it will obtain its base form by removing the word inflectional form and maintaining its base form [21].

One helpful approach to word normalization is Stemming. It is a method that removes words that have the same root as a standard form [21]. It excludes all its derivatives and inflectional affixes from each word that it gets processed with.

Normalization in statistics is the scaling down of datasets so that the normalized data will be between 0 and 1 [22]. The expression below is the formula of normalization:

$$x_{normalized} = \frac{(x - x_{min})}{(x_{max} - x_{min})}$$

The formula first identifies the data set's max and min values, then computes the range by deducting the min from the max value. Then, the value to be normalized from the min will be determined by deducting the min value from variable x. Lastly, it will be divided, and the output is the normalized variable x [22].

2.3.1.3 Punctuation Removal

Another essential part of word normalization is punctuation removal. This method removes any punctuation symbols since it does not help the approach of the researchers in mind. These marks are not useful since they will significantly affect the outcome of the model, depending on how words are frequently used in any sentences or phrases. By removing them, the result

will be more precise and concise in detecting the possible meaning of words throughout each piece of data being fetched [5].

2.3.1.3 Extraction, Categorization, and Feature Selection

This section shows the improvements in the researchers' study on the Bidirectional LSTM algorithm. These layers will help to achieve an improvement in its accuracy in translating from one language to another.

2.3.1.4 Embedding Layers

The Embedding layer is a layer that is hidden in a neural network. It will make word vectors for each word that the model will soon use. The output of the embedding layer is the input of the LSTM layer, which is placed in the middle of each LSTM layer on the Bi-LSTM layer. The model, word level, and embedding vectors will be used to input and handle vocabulary words or OFV [16].

$$y = E * x + b$$

The above-seen formula is the formula of the embedding layer. It shows that a straight-line relationship between each variable, x and y. E decides on which direction or what the words are on the following results to be transfer on the layer and y intercepts b to define where it interacts.

2.3.1.5 Dropout Layers

It verifies and removes any nodes that are present in a neural network. All the forward (future) and the backward (past) are links and drop nodes removed in the interim, resulting in a new branch of the network out of the parent network structure. In training, each iteration will increase the probability of exclusion associated with individual neurons from the neural network, and when at the time of its operation, it will disregard a subset of neurons to lessen overfitting [17].

$$y = \frac{x * mask}{1 - Dp Rate}$$

The formula shows x as an input of the dropout layer. The mask is a binary mask where the elements are randomly selected, either 0 or 1. The DP rate or Dropout rate determines the drop rate during training. It is set at a 1-Dp rate to ensure that the expected value output remains relatively the same.

2.3.1.6 Dense Layers

This layer is used for classification and is the building block of a neural network [18]. Its parameter is used to learn during each training session to produce a more accurate prediction output.

$$Output = act(dot(x, kernel) + bias)$$

The formula shows that the activation function will apply elementwise to the result, which gives a non-linearity to the layer. To get output x is the input which represents the iteration of neuron on each previous layer. Then, weight represents the connection between the iteration of neurons to the current neuron. Lastly, bias were added to the weighted iteration.

2.4 Bidirectional LSTM with Statistical Machine Translation (SMT)

Before there was a translation, it was used to decode and encode transmitted messages during wars. However, as the world began to have peace, the process became translation from one language

to another. SMT, or Statistical Machine Translation, is one of the results of this [6]. It has been used plenty of times to translate one language to another in the early 2010s. It uses a translation model that identifies similarities between the sentence from a source corpus and the target corpus within a source/target parallel corpus [6]. The architecture of STM in Figure 4 can be seen as a formula from Bayes' Theorem [19].

In this study, the researchers will utilize the word-based model that Statistical Machine Translation supports via a toolkit called word alignment. It can process from the source to target words, one-to-zero, one-to-one, and one-to-many [6]. Many studies have used Bi-LSTM with Statistical Machine Translation. The Korean-Vietnamese SMT system was improved using Bi-LSTM, developed by Nguyen et al. [11]. Another significant study by Lo et al. [1] uses SMT for their French-Senegalese SMT system with a base accuracy of 68% in translation. The researchers projected that it could achieve an acceptable accuracy in the Filipino-Hiligaynon SMT system. The researchers now have a baseline to test using a model of an SMT with Bi-LSTM to translate Filipino to Hiligaynon and vice versa. The Tokenization, normalization, and data pre-processing are shown in 2.3, and the researchers also reached the specific objective for data pre-processing using the provided data sets in 2.1.

2.4.1 Language & Translation Models

These models are used by many projects in Statistical Machine Translation. It measures how likely the sentence is generated to be fluent next to a native speaker of the source language. The formula below represents the language model:

$$P(\text{Source}W) = P(\text{source}W_1 \dots \text{source}W_n)$$

The source will find the probability and divide it by the number of words so that the model can get the probability of times that it would be repeated in a corpus. Its other term is the estimate of the occurrence of a sentence. The word-based probability will be used in this study, as represented in the formula above. The $P(\text{source}W)$ is the probability of a whole sequence of words, and the $P(\text{source}W_1)$ will be multiplied by the $P(\text{source}W_2)$, given the $P(\text{source}W_1)$ and so on, multiplied by the probability of the last selected word, given all the previous word chosen. Making sentences more fluent and word probabilities will have histories. SMT system selects the most fluent sentence in the target corpus in translation. [21].

$$Pr(\text{Source}|\text{Target}) = \frac{Pr(\text{Source})Pr(\text{Target}|\text{Source})}{Pr(\text{Target})}$$

The formula shows that every pair is assigned a string, source, and the target corpus in a $Pr(\text{Source}|\text{Target})$. When it is processed, the source corpus will be interpreted, and a probability of translation will be given to produce an acceptable/accurate translation in the target corpus, depending on the training time [19]. The source corpus will observe the target corpus and consider the prior probability of the source getting an accurate translation to the target corpus [19]. The equation of Machine Translation in the statistical model is shown below.

$$\text{Source} = \text{argmax} Pr(\text{Source}) Pr(\text{Target}|\text{Source})$$

2.4.2 Evaluation

Evaluation is essential to a machine translation system to measure its quality. It will increase the process significantly and

make sure that the quality of the translation is acceptable or accurate to use in the real world [21]. Human and Automatic evaluation are the two most reliable evaluation methods in machine translation. The researchers will utilize these two evaluation methods with lexical similarities between the source and target language, Hiligaynon and Filipino, since they are in a related family of languages. The researchers will rely on lexical similarity. The lexical metrics' dependability is determined by the significance of the heterogeneity/representativity of reference translations.

Example:

“Ma-aari kayong pumasok sa ating klase”

Ref1: “Ma-aari na kayong pumasok sa klase”

Ref2: “Pasok na kayo sa ating klase”

Ref3: “Pwede na kayong pumasok sa klase”

Human Evaluation is the golden standard for evaluation since it comes from a native speaker who conveys feelings that result in the subjectivity of the sentence. It makes the translation more authentic but consumes more time and resources and is expensive, unlike Automatic Evaluation, which is cheaper, faster, and uses data to set its parameters. [21]

Bilingual Evaluation Understudy, or its abbreviation BLEU, is a metric used to evaluate the quality of machine translations [25]. It uses reference translation to measure the quality of the trained model, measures the quality of the machine-generated text based on precision, and evaluates the text translation based on matching words (n-grams). BLEU Score outputs a value of 0 and 1 for its evaluation. 0 refers to the machine-translation output that has no overlap from the machine-translated text with the reference translation. In contrast, 1 refers to the exact or perfect translation generated by the machine with the reference translation [26]. The representation of the BLEU score is through percentage. Unigrams and bigrams are used to construct the language model, which aids in learning patterns from the source to the target language. This approach captures sets of words within a specific window, with bigrams set to $n=1$ and unigrams set to $n=2$ [24]. After running simulations using both the baseline and suggested Bi-LTSM models, the researchers obtained the necessary data to create an evaluation between both models, which will be discussed in Chapter 3.

3. RESULTS AND DISCUSSION

3.1 Model Implementation

The base model uses a simple statistical machine translation using the IBM Model 1. It is known for its simplicity and effectiveness in estimating lexical translations from a source to a target language. The model operates on a word alignment mechanism fundamental to SMT. It is initialized using a word alignment technique called Hidden-Markov Alignment, as it is a word-based translation [23]. Data is cleaned, tokenized, and processed to build a vocabulary essential for the model's training phase. Once the data has been pre-processed and ready to be received by the model, the data goes through each gate of the Bi-LSTM model, which is shown in Chapter 2.2, where the gates are presented mathematically. Each gate processes the data in forward and backward directions and concatenates the results. Once the Bi-LSTM finishes the translation, it is sent to the translation model to create the intended output. The process also allows the researchers to get probability and BLEU scores from each translation. Both scores help with determining the quality of the outputs; the researchers measure translatability and error

frequency using the probability score as a metric if the data is translated, mistranslated, or untranslatable, along with the quality of the translation, if the translation was equal, under translated, over translated, or complete as seen in Figures 10 and 11.

3.2 Training Process

The training process involves the iterations of each tokenized word to estimate the probabilities of each word, with the condition set for valid pair *count (source_word, target_word) > 0*. Unigrams and bigrams are the basis of the creation of the language structure. It facilitates the pattern learning process from the source language to the target language as it is a set of occurring words within a given window, such as bigram n=1 and unigram n=2 [24]. Furthermore, it splits sentences into small groups of words to understand their meaning and context. The researchers’ datasets consist of various words and sentences of target and source languages, but they are limited to less than 5000 unique words and 2000 words.

3.3 Base Model Results Evaluation

The model uses the BLEU score to evaluate the quality of translation, a leading metric that measures machine translation performance, such as Statistical Machine Translation. The BLEU score ranges between zero and one, where a higher score indicates better translation quality. The results of the Base Model showed a BLEU score of **24.81**, as illustrated in **Figure 6**

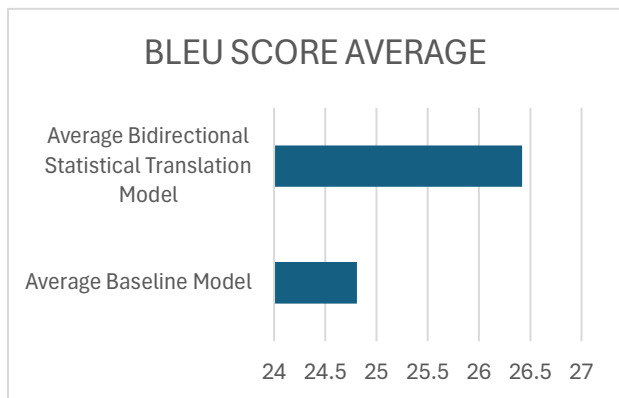


Figure 6. Bleu Score Average

Indicating basic understanding but with notable grammatical inaccuracies.

The Base Model results in the average probability score, obtaining a confidence level of **55.35%**, as shown in **Figure 7**

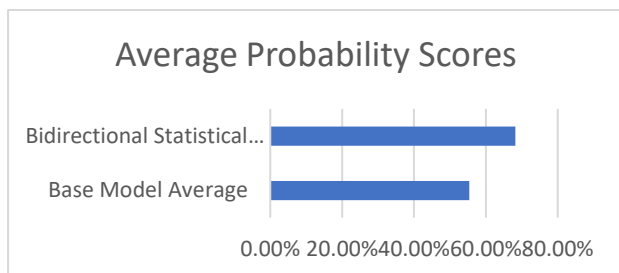


Figure 7. Average Probability Score

This score represents the likelihood that the Base Model assigns to its predicted translations being correct. Furthermore, the

researchers plotted the probability score distribution across multiple translation samples, presented in **Figure 9**

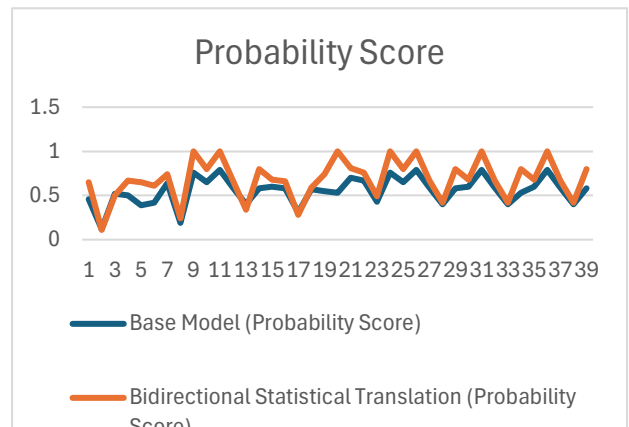


Figure 9. Probability Score

It is noticeable that the Base Model’s probability scores fluctuate, suggesting inconsistencies in translation confidence. The error frequency analysis for both Tagalog and Hiligaynon translations, presented in **Figure 12**

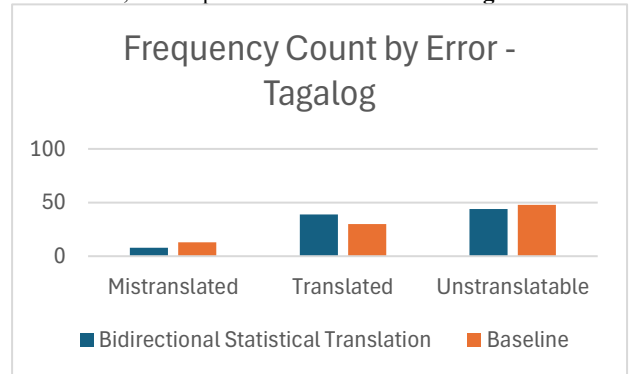


Figure 12. Frequency Count by Error - Tagalog

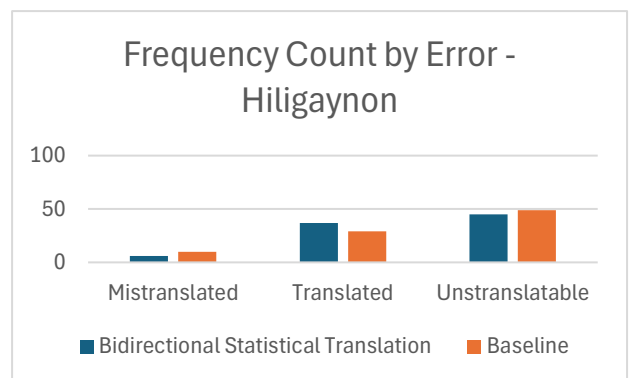


Figure 13. Frequency Count by Error - Hiligaynon

The results show that the Base Model had a considerable number of mistranslated words and a high number of untranslatable words for both languages. It indicates the model’s difficulty in handling complex vocabulary and sentence structures. The high frequency of mistranslations and untranslatable words highlights the Base Model’s limitations in accurately translating between Filipino and Hiligaynon.

3.4 Enhancement with Bi-LSTM

Integrating a Bi-LSTM network into the translation model improved the translation accuracy, achieving a BLEU score of **26.42**, representing a **1.61% increase**; **Figure 6** shows the data below.

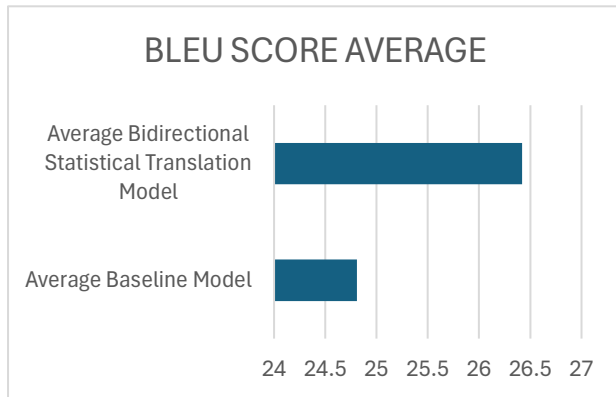


Figure 6. Bleu Score Average

Bi-LSTM’s dual-directional understanding improved, enhancing the depth of sequence processing and context comprehension.

The Bidirectional Statistical Translation Model also showed a significant improvement in the average probability score, increasing by **12.9%** compared to the Base Model, as illustrated in **Figure 7**

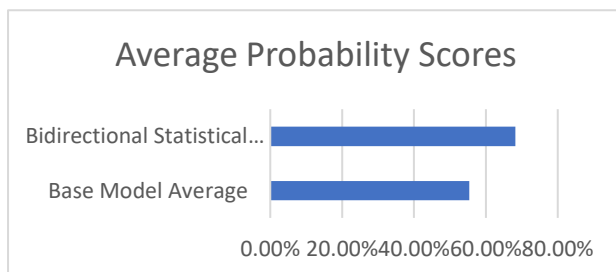


Figure 7. Average Probability Score

This increase indicates greater confidence in the translation outputs, supported by additional layers like Embedding and Dense layers, which refine the model’s understanding of context.

The probability score distribution for the Bidirectional Statistical Translation Model, displayed in **Figure 9**

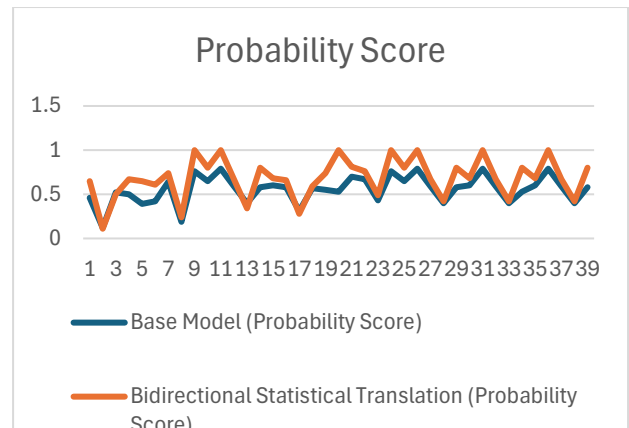


Figure 9. Probability Score

It demonstrates more consistent and higher confidence levels across various samples. This stability suggests that the model’s enhancements allow it to provide more reliable translation outputs.

The combined error frequency analysis for both Tagalog and Hiligaynon, presented in **Figure 12**,

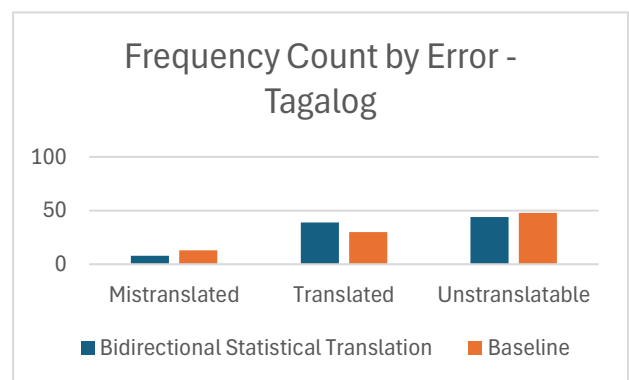


Figure 12. Frequency Count by Error - Tagalog

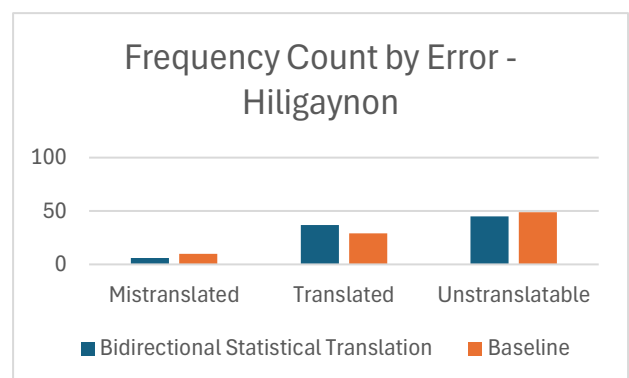


Figure 13. Frequency Count by Error - Hiligaynon

highlights the improvements made by the Bidirectional Statistical Translation Model. It significantly reduced the number of mistranslated words, and there was an increase in the count of successfully translated words for both languages. Additionally, the frequency of untranslatable words decreased, indicating better handling of complex and rare vocabulary. These results demonstrate the effectiveness of the enhancements,

particularly the integration of Bi-LSTM, in improving the model's translation capabilities and reducing errors.

4. CONCLUSION AND RECOMMENDATION

The enhancement with Bi-LSTM improved the statistical machine translation and highlighted further potential areas for refinement and advancement in the system.

4.1 Conclusions

This study demonstrated the potential of integrating Bidirectional Long Short-Term Memory (Bi-LSTM) into a Statistical Machine Translation (SMT) framework to enhance translation accuracy between Filipino and Hiligaynon. The Bidirectional Statistical Translation Bi-LSTM model significantly improved baseline accuracy, with BLEU scores indicating an upward trend in translation quality. The forward-backward processing capabilities of Bi-LSTM facilitated a more nuanced handling of language context, which is essential in capturing the subtle contextual distinctions between Filipino and Hiligaynon.

Going into the specifics, the IBM Model 1 Statistical Machine Translation achieved a BLEU score of 24.81%, with a fair translation quality and noticeable yet passable grammatical errors. Integrating Bi-LSTM improves the translation accuracy to 26.42%, demonstrating the effectiveness of bidirectional contextual learning. The model provides a solid foundation for word-based translation and a probabilistic translation model, and the addition of Bidirectional LSTM enhanced its translation quality by incorporating multiple layers and its capability of contextualizing data from both directions. The results confirm that Bi-LSTM-enhanced SMT models can effectively bridge the gaps between languages regardless of lexical and semantic complexity.

4.2 Recommendation

In this study, the researchers have several recommendations. First, assessing translation requirements could clarify whether to optimize IBM Model 1 further or transition to more robust IBM models for specific language pairs. Comparative studies should evaluate enhanced SMT models like Bi-LSTM against modern models such as Transformers, measuring accuracy, processing time, and adaptability across informal and formal texts. Given the superior contextual capabilities of Neural Machine Translation (NMT) systems, particularly with Bi-LSTM integration, transitioning to NMT is advisable. Expanding the Dataset is recommended, especially for less-resourced languages like Hiligaynon, to reduce errors and improve translation accuracy. If any problems occur with the application program, creating a web app can also be a viable option. The researchers recommend using a microphone for input and a speaker output as an accessibility feature that enables interactions using voice inputs.

5. REFERENCES

- [1] A. Lo, C. B. Dione, E. M. Nguer, S. Ba, and M. Lo, "Using LSTM Network to translate French to Senegesele local languages: Wolof as a case study," Thesis, University of Berger, Universite Virtuelle du Senegal, 2016. [Online]. Available: <https://arxiv.org/pdf/2004.13840.pdf>
- [2] N. Bensalah, H. Ayad, A. Adib, and A.I. El Faroul, *LSTM vs GRU for Arabic Machine Translation*, vol. 1383. Advance in Intelligent Systems and Computing. [Online]. Available: https://doi.org/10.1007/978-3-030-73689-7_16
- [3] M. Bowman, S. Debray, and L. Peterson, "Reasoning About Naming Systems," The Pennsylvania State University, The University of Arizona, 1993. [Online]. Available: <http://doi.acm.org/10.1145/161468.161471>
- [4] L. D. Brown, H. Hua, and C. Gao, *A widget framework for augmented interaction in SCAPE 2003*. In Proceedings of the 16th Annual ACM Symposium on User interface Software and Technology, 2003, pp. 1–10. doi: 10.1145/964696.964697.
- [5] H. Elfaik and E. H. Nfaoui, "Deep bidirectional LSTM Network Learning-Based Sentiment Analysis for Arabic text," *Journal of Intelligent Systems*, vol. 30, no. 1, pp. 395-412, Dec. 2020, doi: 10.1515/jisys-2020-0021.
- [6] H. T.-T. Do. Huynh, K. V. Nguyen, N. L.-T. Nguyen, and A. G.-T. Nguyen, "Hate Speech Detection on Vietnamese Social Media Text using the Bidirectional-LSTM Model" University of Information Technology, 2019. [Online]. Available: <https://arxiv.org/pdf/1911.03648>
- [7] V. Makarencov, L. Rokach, and B. Shapira, *Choosing the right word: Using bidirectional LSTM tagger for writing support systems*. Engineering Applications of Artificial Intelligence, vol. 84, 2019, 1–10. <https://doi.org/10.1016/j.engappai.2019.05.003>
- [8] H. Mohaouchane, A. Mourhir, and N. Nikolov, "Detecting offensive language on Arabic social media using deep learning," *IEEE Conference Publication | IEEE Xplore*, 2019. <https://ieeexplore.ieee.org/document/8931839>
- [9] M. K. Negia, R. M. Tamiru, and M. Meshesha, "Amrahic-Kistanigna Bidirectional Machine Translation using Deep Learning," *IEEE Conference Publication | IEEE Xplore*, 2022. <https://ieeexplore.ieee.org/document/9971194>
- [10] C. Yichao, F. Yang, Q. Tang and X. Lu, "An Attention Enhanced Bidirectional LSTM for Early Forest Fire Smoke Recognition," 2019. <https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=8865046>
- [11] V.-H. Vu, Q.-P. Nguyen, K.-H. Nguyen, J.-C. Shin, and C.-Y. Ock, "Korean-Vietnamese Neural Machine Translation with Named Entity Recognition and Part-of-Speech Tags," *IEICE Transactions on Information and Systems*, vol. E103. D, no 4, pp. 866-873, 2020, doi: 10.1587/transinf.2019edp7154.
- [12] Y.T. Yu, M.F. Lau, "A comparison of MC/DC, MUMCUT and several other coverage criteria for logical decisions," *Journal of Systems and Software*, 2005, in press.
- [13] Z. Huang, P. Xu, D. Liang, A. Mishra, & B. Xiang, "TRANS-BLSTM: Transformer with Bidirectional LSTM for Language Understanding." 2020 ArXiv: Computation and Language.
- [14] "What is Tokenization?" *IXOPAY*, Oct. 01, 2024. [Online] Available: <https://www.tokenex.com/blog/what-is-tokenization/>
- [15] T. Duque, "Text normalization," *Towards Data Science*, Mar. 05, 2025. [Online] Available:

<https://towardsdatascience.com/text-normalization-for-natural-language-processing-nlp-70a314bfa646>

- [16] C. Minsoo, H. Jihwan, P. Chihyun, P Sanghyun, "Combinatorial Feature embedding based on CNN and LSTM for biomedical named entity recognition," Yonsei University, 2020 [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1532046420300083>
- [17] A. Labach, H. Salehinejad, S. Valaee, "Survey of Dropout Methods for Deep Neural Networks." University of Toronto, 2019. [Online]. Available: <https://arxiv.org/pdf/1904.13310.pdf>
- [18] V. L. H. Josephine, A. P. Nirmala, and V. L. Alluri, "Impact of Hidden Dense Layers in Convolutional Neural Network to enhance Performance of Classification Model," IOP Conference Series Materials Science and Engineering, vol. 1131, no. 1, p. 012007, Apr. 2021, doi: 10.1088/1757-899x/1131/1/012007
- [19] P. F. Brown, V. J. Della Pietra, S. A. Della Pietra, and R. L. Mercer, "The mathematics of statistical machine translation: parameter estimation," Computational Linguistics, vol. 19, no. 2, pp. 263-311, Jun. 1993, [Online]. Available: <https://aclanthology.org/J93-2003.pdf>
- [20] Z. Tan, Et al. Neural machine translation: A review of method, resources, and tools, AI Open. [Online] Available: <https://www.sciencedirect.com/science/article/pii/S2666651020300024>
- [21] C. Bonet, "Statistical Machine Translation A practical Tutorial," Universitat Politècnica de Catalunya. <https://www.cs.upc.edu/~cristinae/CV/docs/tutorialSMTprint.pdf>
- [22] W. Team, "Normalization Formula," What Is It, How To Calculate, 2025. <https://www.wallstreetmojo.com/normalization-formula/>
- [23] F. J. Och and H. Ney, "Improved Statistical Alignment Models." University of Technology. [Online]. Available: <https://aclanthology.org/P00-1056.pdf>
- [24] Oorja, "what is the difference between bigram and unigram text feature extraction," Stackoverflow, 2017. <https://stackoverflow.com/questions/43463792/what-is-the-difference-between-bigram-and-unigram-text-features-extraction>
- [25] H. Tran, H. Tran, S. Nguyen, H. Nguyen, & T. Nguyen, "Does BLEU Score work for Code Migration?," IEEE Conference Publication | IEEE Xplore, May 01, 2019. <https://ieeexplore.ieee.org/document/8813269>.
- [26] Google Cloud, "Evaluating Model, Understanding the BLEU Score". Guides. [https://cloud.google.com/translate/automl/docs/evaluate#:~:text=BLEU%20\(BiLingual%20Evaluation%20Understudy\)%20is,of%20high%20quality%20reference%20translations.](https://cloud.google.com/translate/automl/docs/evaluate#:~:text=BLEU%20(BiLingual%20Evaluation%20Understudy)%20is,of%20high%20quality%20reference%20translations.)
- [27] Alex, "Statistical vs Neural Machine Translation | Prestige Network," Prestige Network, Jan. 18, 2024. <https://www.prestigenetwork.com/insights/the-difference-between-statistical-and-neural-machine-translation/>
- [28] C. L. Motus, Hiligaynon Dictionary, PALI Language Texts-Philippines. <https://doi.org/10.2307/j.ctv9hvsxq>
- [29] "UP diksiyonaryong Filipino" Internet Archive, 2001. <https://archive.org/details/updiksiyonaryong000unse>

APPENDIX A

Authorization Letter

This letter authorizes **MR. RAMCIS N. VILCHEZ**, the adviser and co-author of the study entitled **A DEEP LEARNING-ENHANCED STATISTICAL MACHINE TRANSLATION OF FILIPINO LOCAL LANGUAGES**, to present the study at local, national, or international research conference; to publish the paper in local, national, or international research journals; and/or apply for national or international intellectual property protections. Moreover, the adviser ensures that the primary authors receive due credit.

The Researchers

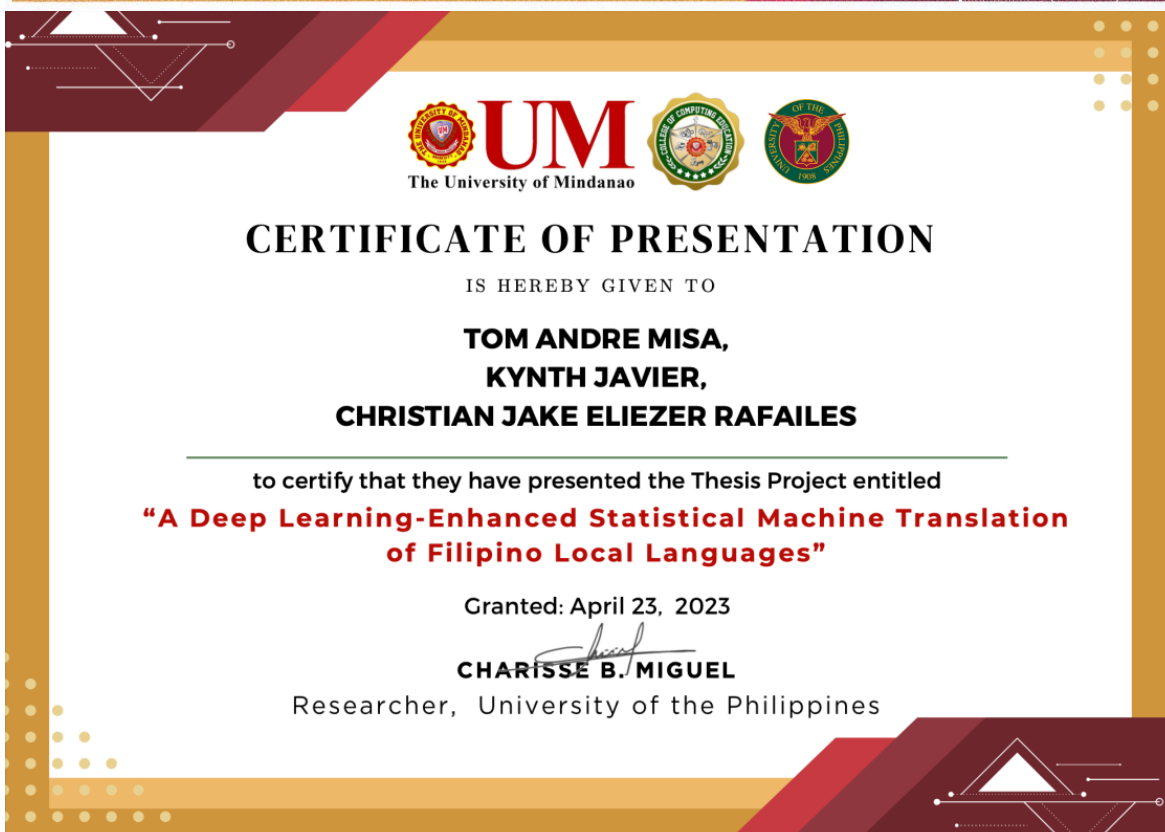
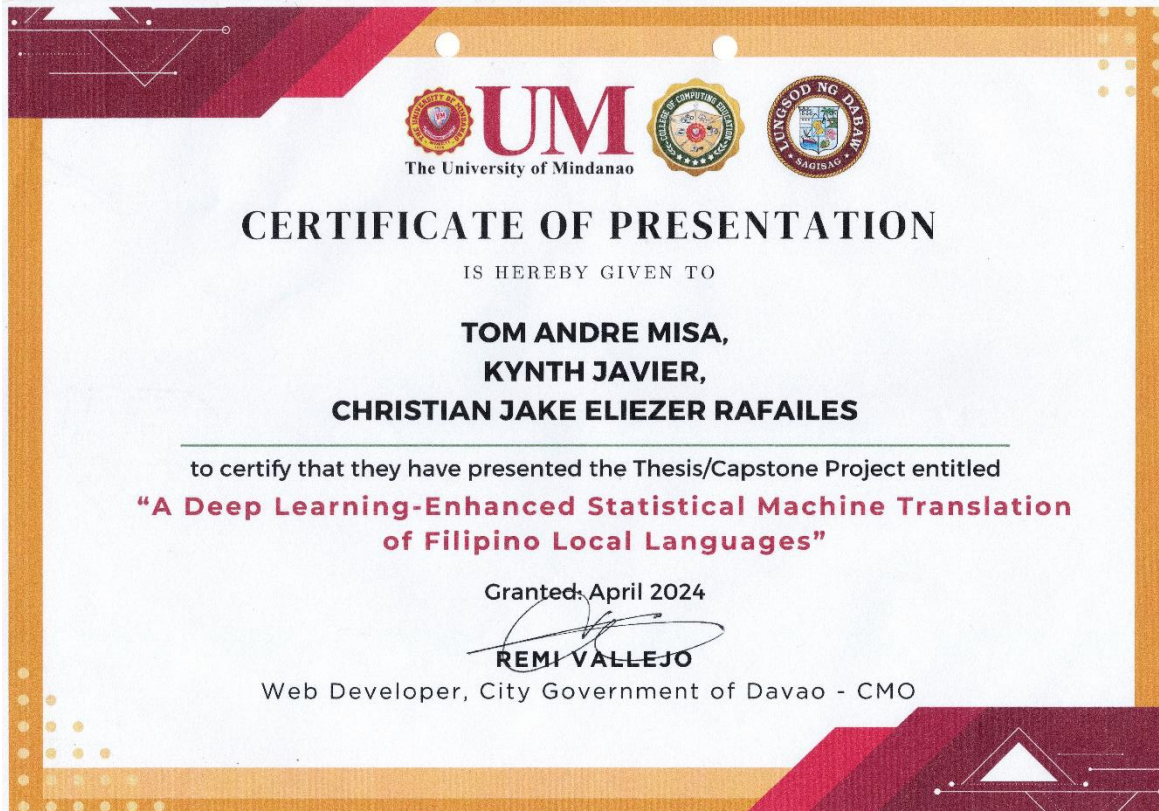
KYNTH I. JAVIER

TOM ANDRE B. MISA

CHRISTIAN JAKE ELIEZER B. RAFAILES

APPENDIX B

Certificate of Presentation





UM

The University of Mindanao



CERTIFICATE OF PRESENTATION

IS HEREBY GIVEN TO

**TOM ANDRE MISA,
KYNTH JAVIER,
CHRISTIAN JAKE ELIEZER RAFAILES**

to certify that they have presented the Thesis/Capstone Project entitled
**"A Deep Learning-Enhanced Statistical Machine Translation
of Filipino Local Languages"**

Granted: April 23, 2023

JAYNARD JOHN HERNANDO
Professor, University of Mindanao

Certificate of Acceptance

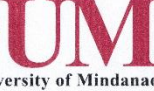


CERTIFICATE OF ACCEPTANCE
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**TOM ANDRE MISA,
KYNTH JAVIER,
CHRISTIAN JAKE ELIEZER RAFAILES**

to certify that they have presented the Thesis Project entitled
**"A Deep Learning-Enhanced Statistical Machine Translation
of Filipino Local Languages"**

Granted: April 23, 2023


CHARISSE B. MIGUEL
Researcher, University of the Philippines



CERTIFICATE OF ACCEPTANCE
IS HEREBY GIVEN TO
**TOM ANDRE MISA,
KYNTH JAVIER,
CHRISTIAN JAKE ELIEZER RAFAILES**

to certify that the Capstone Project/Thesis entitled
**"A Deep Learning-Enhanced Statistical Machine Translation
of Filipino Local Languages"**
is accepted and to be utilized by the company.

Granted: April 23, 2023


JAYNARD JOHN HERNANDO
Professor, University of Mindanao



The University of Mindanao

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CERTIFICATE OF ACCEPTANCE

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**TOM ANDRE MISA,
KYNTH JAVIER,
CHRISTIAN JAKE ELIEZER RAFAILES**

to certify that the Capstone Project/Thesis entitled
**"A Deep Learning-Enhanced Statistical Machine Translation
of Filipino Local Languages"**
is accepted and to be utilized by the company.

Granted: April 2024

REMI VALLEJO

Web Developer, City Government of Davao - CMO

Certificate of Public Presentation



APPENDIX C

Endorsement Letter for Final Defense



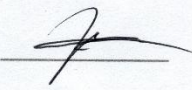
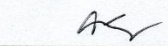
College of Computing Education
Office of the Research Coordinator
3rd Flr, DPT Building, Matina Campus, Davao City
Tel No. (082) 300.5456 loc 192

04/20/24

Endorsement for Final Defense

This is to endorse the **Thesis** entitled **A Deep Learning-Enhanced Statistical Machine Translation of Filipino Local Languages** which were prepared and submitted by **Tom Andre Misa, Christian Jake Eliezer Rafalles, Kynth Javier** for FINAL Defense.

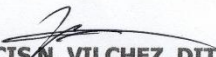
The paper has been evaluated by the research personnel listed below and was found to be in compliant with the quality standards and policies stipulated in the UM Research Manual.

	Name	Signature
Adviser	Ramcis N. Vilchez, DIT	
Subject Teacher	Charisse P. Barbosa, MSIS	

Noted by:


CHARISSE P. BARBOSA, MSIS
Research Coordinator, CCE

Approved by:


RAMCIS N. VILCHEZ, DIT
Dean, CCE

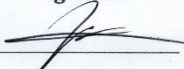
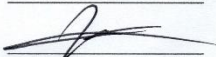
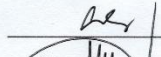
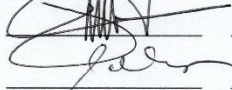
Assignment of Research Personnel

 UM The University of Mindanao	RESEARCH AND PUBLICATION CENTER [x] Main [] Branch _____ ASSIGNMENT OF RESEARCH PERSONNEL
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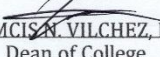
Course Code: BSM 325

Program: BS Computer Science

This is to acknowledge acceptance of assignment as Research Personnel for the thesis entitled:
A DEEP LEARNING-ENHANCED STATISTICAL MACHINE TRANSLATION OF FILIPINO LOCAL
LANGUAGES

	Name	Signature	Date
Adviser	<u>RAMCIS N. VILCHEZ, DIT</u>		_____
Statistician/Data Analyst	_____	_____	_____
Editor	<u>RAMCIS N. VILCHEZ, DIT</u>		_____
Panel Members	<u>CHARISSE P. BARBOSA, MSIS</u>		_____
	<u>MELJOHN V. ABORDE, MIT</u>		_____
	<u>RANDY F. ARDEN</u>		_____

Endorsed by:  FE B. YARA, MSIS
Research Coordinator/Asst. Research Coordinator

Approved by:  RAMCIS N. VILCHEZ, DIT
Dean of College

APPENDIX D

Final Defense Evaluation Forms

 UM The University of Mindanao	RESEARCH AND PUBLICATION CENTER [X] Main [] Branch _____ THESIS & CAPSTONE II (Evaluation Form)
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Title: A Deep Learning Enhanced Statistical Machine Translation of Filipino Local Languages

Proponents: Tom Andre Mira
Christian Jake Papailso
Kyuth Javier

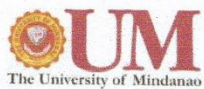
Instructions:

Enclosed are the criteria-indicators and rating scale to help you evaluate the various components of this thesis work. Please read the criteria carefully and place the corresponding score of each item on the blank provided for. Multiply the score with its corresponding weight and enter it on last column. Add the points in the last column to compute the final grade. Space is provided for your comments at the bottom of the page. Thank you.

Rating Scale:

Score	Equivalent	Narrative Description
100.00 - 95.00	Outstanding	The indicators of the criterion are carried out in an excellent manner.
94.99 - 90.00	Very Good	The indicators of the criterion are carried out evidently.
89.99 - 85.00	Good	The indicators of the criterion are carried out fairly.
84.99 - 80.00	Poor	The indicators of the criterion are carried out deficiently.
79.99 - 75.00	Very Poor	The indicators of the criterion are carried out very deficiently.
74.99 - 70.00	Missing	The indicators of the criterion are not evident at all.
N/A		Not Applicable

Criterion	Score	Weight	Points
Introduction An exposition on the Project Context, Purpose and Description of the study are given. The objectives are specific, measurable, achievable, realistic and time-bound. The scope and delimitations of the study are clearly explained.		25%	21
Related Literature There is evidence that the writer made a scholarly attempt to find his precursors in the field. The review provides relevant discussion on theories, concepts, facts, and/or ideas related to the subject. The materials are current, adequate and carefully selected. A Synthesis is provided at the end of the section describing the review's significance.		10%	9



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THESIS & CAPSTONE II (Evaluation Form)

Criterion	Score	Weight	Points
General Format The manuscript is very well-organized and contains all the main requisites of a typical thesis manuscript. The prescribed rules on the format, spacing, pagination, tables and figures are strictly followed. Mechanics of writing is satisfactory. The style is impersonal, straight forward, objective and effective.		15%	13
Oral Presentation Oral presentation is methodical, rational and coherent. It is brief but complete. Appropriate and attractive visual aids are used. Presenters are articulate and questions are answered satisfactorily.		15%	13
System Software • Punctuality • Reliability • Usability • Efficiency • Maintainability • Portability		35%	29
TOTAL		100%	85 mg ✓

Comments:

Recommendation:

The paper is: [] Accepted

[] Accepted with Minor Revision

[x] Accepted with Major Revision

[] Rejected


CHARISSE P. BARBOSA

Name and Signature of the Panel Member

7/7/2025
Date



RESEARCH AND PUBLICATION CENTER

[X] Main [] Branch

THESIS & CAPSTONE II (Evaluation Form)

Title: A Deep Learning Enhanced Statistical Machine Translation of
Filipino Local Language

Proponents: Don Andre Misa
Christian Dale Repaila
Kynthe Javier

Instructions:

Enclosed are the criteria-indicators and rating scale to help you evaluate the various components of this thesis work. Please read the criteria carefully and place the corresponding score of each item on the blank provided for. Multiply the score with its corresponding weight and enter it on last column. Add the points in the last column to compute the final grade. Space is provided for your comments at the bottom of the page. Thank you.

Rating Scale:

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74.99 - 70.00	Missing	The indicators of the criterion are not evident at all.
N/A		Not Applicable

Criterion	Score	Weight	Points
Introduction An exposition on the Project Context, Purpose and Description of the study are given. The objectives are specific, measurable, achievable, realistic and time-bound. The scope and delimitations of the study are clearly explained.		25%	22
Related Literature There is evidence that the writer made a scholarly attempt to find his precursors in the field. The review provides relevant discussion on theories, concepts, facts, and/or ideas related to the subject. The materials are current, adequate and carefully selected. A Synthesis is provided at the end of the section describing the review's significance.		10%	7



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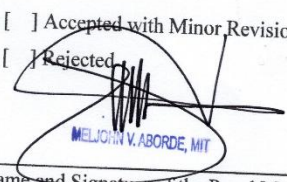
Criterion	Score	Weight	Points
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Oral Presentation Oral presentation is methodical, rational and coherent. It is brief but complete. Appropriate and attractive visual aids are used. Presenters are articulate and questions are answered satisfactorily.		15%	13
System Software • Punctuality • Reliability • Usability • Efficiency • Maintainability • Portability		35 %	21
TOTAL		100%	47

Comments:

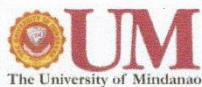
Recommendation:

The paper is: ☐ Accepted
☒ Accepted with Major Revision

☐ Accepted with Minor Revision
☒ Rejected


Name and Signature of the Panel Member

Date



RESEARCH AND PUBLICATION CENTER

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THESIS & CAPSTONE II (Evaluation Form)

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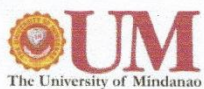
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94.99 - 90.00	Very Good	The indicators of the criterion are carried out evidently.
89.99 - 85.00	Good	The indicators of the criterion are carried out fairly.
84.99 - 80.00	Poor	The indicators of the criterion are carried out deficiently.
79.99 - 75.00	Very Poor	The indicators of the criterion are carried out very deficiently.
74.99 - 70.00	Missing	The indicators of the criterion are not evident at all.
N/A		Not Applicable

Criterion	Score	Weight	Points
Introduction An exposition on the Project Context, Purpose and Description of the study are given. The objectives are specific, measurable, achievable, realistic and time-bound. The scope and delimitations of the study are clearly explained.		25%	22
Related Literature There is evidence that the writer made a scholarly attempt to find his precursors in the field. The review provides relevant discussion on theories, concepts, facts, and/or ideas related to the subject. The materials are current, adequate and carefully selected. A Synthesis is provided at the end of the section describing the review's significance.		10%	9



RESEARCH AND PUBLICATION CENTER

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THESIS & CAPSTONE II (Evaluation Form)

Criterion	Score	Weight	Points
General Format The manuscript is very well-organized and contains all the main requisites of a typical thesis manuscript. The prescribed rules on the format, spacing, pagination, tables and figures are strictly followed. Mechanics of writing is satisfactory. The style is impersonal, straight forward, objective and effective.		15%	13
Oral Presentation Oral presentation is methodical, rational and coherent. It is brief but complete. Appropriate and attractive visual aids are used. Presenters are articulate and questions are answered satisfactorily.		15%	13
System Software • Punctuality • Reliability • Usability • Efficiency • Maintainability • Portability		35 %	30
TOTAL		100%	87% ✓

Comments:

Recommendation:

The paper is: [] Accepted [] Accepted with Minor Revision
[x] Accepted with Major Revision [] Rejected


RANDY F. ARUETA
Name and Signature of the Panel Member

7.9.2025
Date

APPENDIX E

Validator's Checklist and Criteria Forms

APPENDIX F

Alpha and Beta Testing Criteria Forms



COLLEGE OF COMPUTING EDUCATION

[x] Main [] Branch _____

ALPHA TESTING CRITERIA

(Final System / Adviser)

Thesis / Capstone Title:

A Deep Learning - Enhanced Statistical Machine Translation of
Filipino Local Languages

Group Members:

Javier, Kynth I.

Misa, Tom Andre B.

Rafales, Christign Jake Eliezer B.

Product Quality:	Weight (100%)	Score
a. Reliability, proper functioning as specified and expected.	10%	10
b. Robustness, acceptable response to unusual inputs, loads and conditions.	10%	10
Functional Testing: execute each use case, use case flow, or function, using valid and invalid data with an objective to verify that:		
a. The expected results occur when valid data is used.	10%	10
b. The appropriate error / warning message are displayed when invalid data is used.	10%	9
Functionality: - Suitability - Accuracy - Interoperability - Compliance - Security	10%	9
Reliability: - Maturity - Fault-tolerance - Recoverability	10%	9
Usability: - Understandability - Learnability - Operability	10%	9
Efficiency: - Time Behaviour - Resource Behaviour	10%	10
Maintainability: A set of attributes that bear on the effort needed to make specified modified modifications.	10%	10
Portability: A set of attributes that bear on the ability of software to be transferred from one environment to another.	10%	10

TOTAL

100%

96

Tester:

[Signature]

Signature over Printed Name

**COLLEGE OF COMPUTING EDUCATION**

[X] Main [] Branch _____

BETA TESTING CRITERIA

(Final System / End-user)

Thesis / Capstone Title:

A DEEP LEARNING - ENHANCED STATISTICAL MACHINE
TRANSLATION OF FILIPINO LOCAL LANGUAGES

Group Members:

JAVIER, KYNTH I. MISA, TOM ANDRE B.
RAFALES, CHRISTIAN JAKE ELIEZER B.

Product Quality:	Weight (100%)	Score
a. Reliability, proper functioning as specified and expected.	5%	5
b. Robustness, acceptable response to unusual inputs, loads and conditions.	5%	5
c. Efficiency of use by the frequent users.	5%	4
d. Easy to use even for the frequent users.	5%	5
Functional Testing: execute each use case, use case flow, or function, using valid and invalid data with an objective to verify that:		
a. The expected results occur when valid data is used.	10%	10
b. The appropriate error / warning message are displayed when invalid data is used.	10%	
Functionality: • Suitability • Accuracy • Interoperability • Compliance • Security	10%	10
Reliability: • Maturity • Fault-tolerance • Recoverability	10%	10
Usability: • Understandability • Learnability • Operability	10%	10
Efficiency: • Time Behaviour • Resource Behaviour	10%	8
Maintainability: A set of attributes that bear on the effort needed to make specified modified modifications.	10%	10
Portability: A set of attributes that bear on the ability of software to be transferred from one environment to another.	10%	10

TOTAL 100% 97The software is easy to use and navigate even for first-time users.
However, efficiency can still be improved.

Tester:

CHARISSE MIGUEL

Signature over Printed Name

**COLLEGE OF COMPUTING EDUCATION**

[X] Main [] Branch _____

BETA TESTING CRITERIA

(Final System / End-user)

Thesis / Capstone Title:

A DEEP LEARNING-EMPOWERED STATISTICAL MACHINE
TRANSLATION OF FILIPINO LOCAL LANGUAGES

Group Members:

JAVIER, KYNTH I. MISA, TOM ANDRE B.
RAFALES, CHRISTIAN JAKE ELIEZER B.

Product Quality:	Weight (100%)	Score
a. Reliability, proper functioning as specified and expected.	5%	4
b. Robustness, acceptable response to unusual inputs, loads and conditions.	5%	4
c. Efficiency of use by the frequent users.	5%	5
d. Easy to use even for the frequent users.	5%	5
Functional Testing: execute each use case, use case flow, or function, using valid and invalid data with an objective to verify that:		
a. The expected results occur when valid data is used.	10%	10
b. The appropriate error / warning message are displayed when invalid data is used.	10%	10
Functionality: - Suitability - Accuracy - Interoperability - Compliance - Security	10%	9
Reliability: - Maturity - Fault-tolerance - Recoverability	10%	9
Usability: - Understandability - Learnability - Operability	10%	9
Efficiency: - Time Behaviour - Resource Behaviour	10%	10
Maintainability: A set of attributes that bear on the effort needed to make specified modified modifications.	10%	10
Portability: A set of attributes that bear on the ability of software to be transferred from one environment to another.	10%	10

TOTAL 100% 95%

Tester:

JAY ARD
Signature over Printed Name

**COLLEGE OF COMPUTING EDUCATION**

[X] Main [] Branch _____

BETA TESTING CRITERIA

(Final System / End-user)

Thesis / Capstone Title:

A DEEP LEARNING-ENHANCED STATISTICAL MACHINE
TRANSLATION OF FILIPINO LANGUAGES

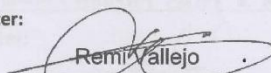
Group Members:

TOM ANDRE MISA, KYNTH JAVIER, CHRISTIAN JAKE
ELIEZER RAFAEL

Product Quality:	Weight (100%)	Score
a. Reliability, proper functioning as specified and expected.	5%	5
b. Robustness, acceptable response to unusual inputs, loads and conditions.	5%	5
c. Efficiency of use by the frequent users.	5%	5
d. Easy to use even for the frequent users.	5%	5
Functional Testing: execute each use case, use case flow, or function, using valid and invalid data with an objective to verify that:		
a. The expected results occur when valid data is used.	10%	9
b. The appropriate error / warning message are displayed when invalid data is used.	10%	9
Functionality: - Suitability - Accuracy - Interoperability - Compliance - Security	10%	10
Reliability: - Maturity - Fault-tolerance - Recoverability	10%	10
Usability: - Understandability - Learnability - Operability	10%	9
Efficiency: - Time Behaviour - Resource Behaviour	10%	8
Maintainability: A set of attributes that bear on the effort needed to make specified modified modifications.	10%	10
Portability: A set of attributes that bear on the ability of software to be transferred from one environment to another.	10%	10

TOTAL **100%** **95 %**

Tester:


Signature over Printed Name

APPENDIX G

Other Figures, Forms, and Tables

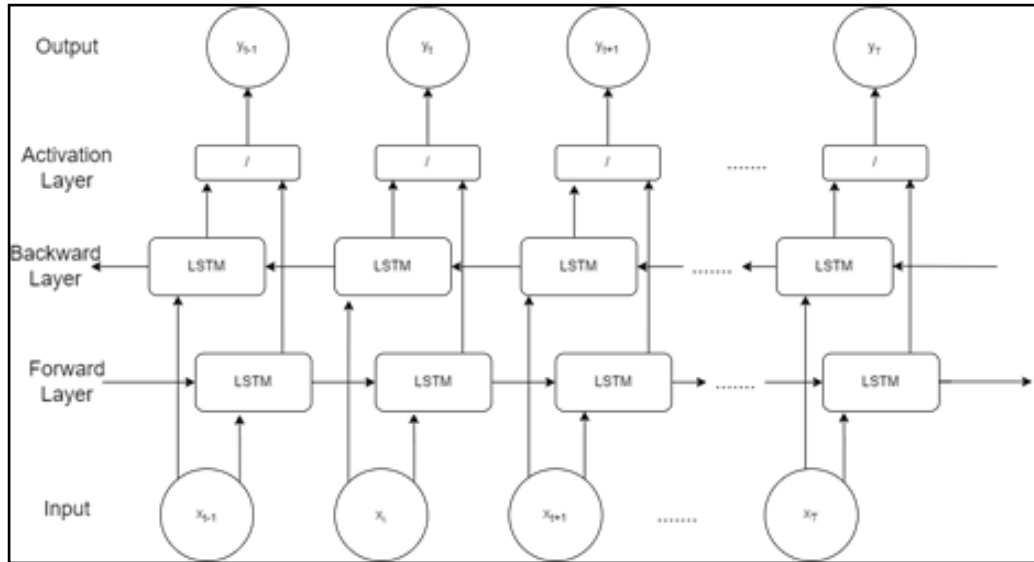


Figure 1. Baseline Bidirectional LSTM

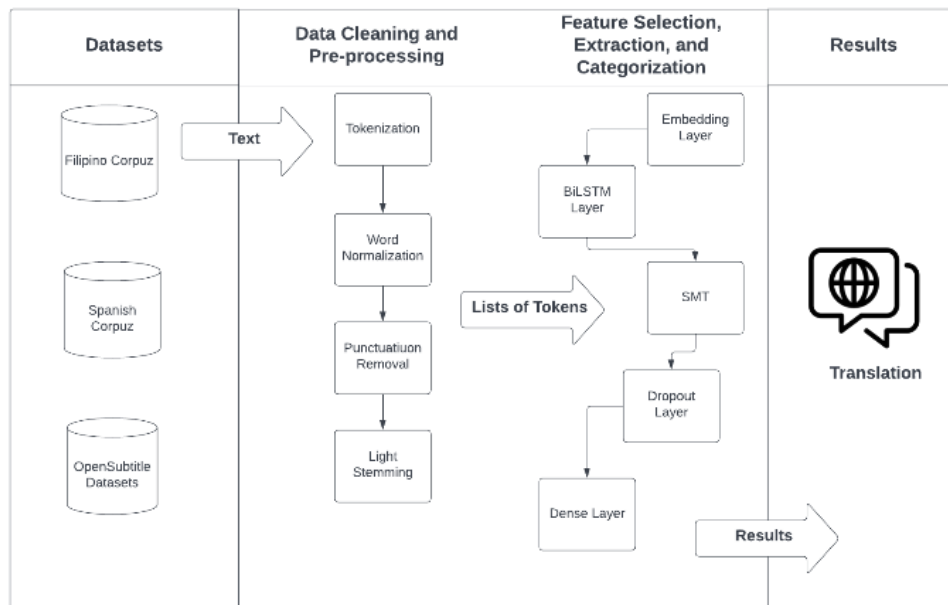


Figure 2. Overall Framework of the Study Bidirectional LSTM

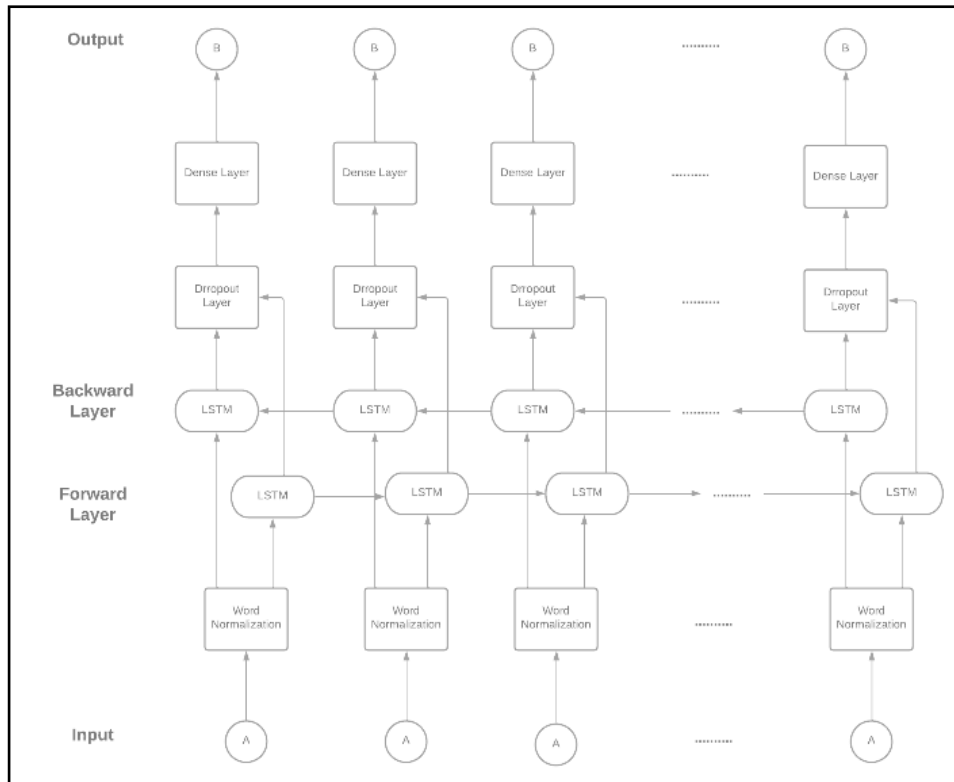


Figure 3. Study Bidirectional LSTM

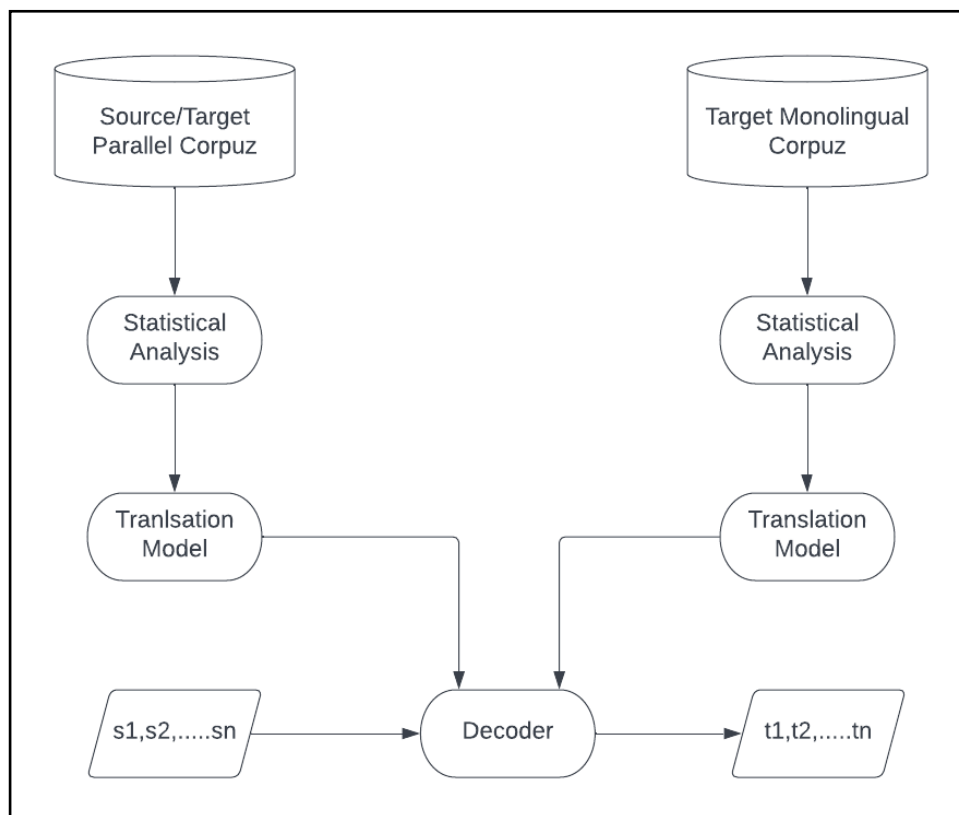


Figure 4. Statistical Machine Translation Architecture

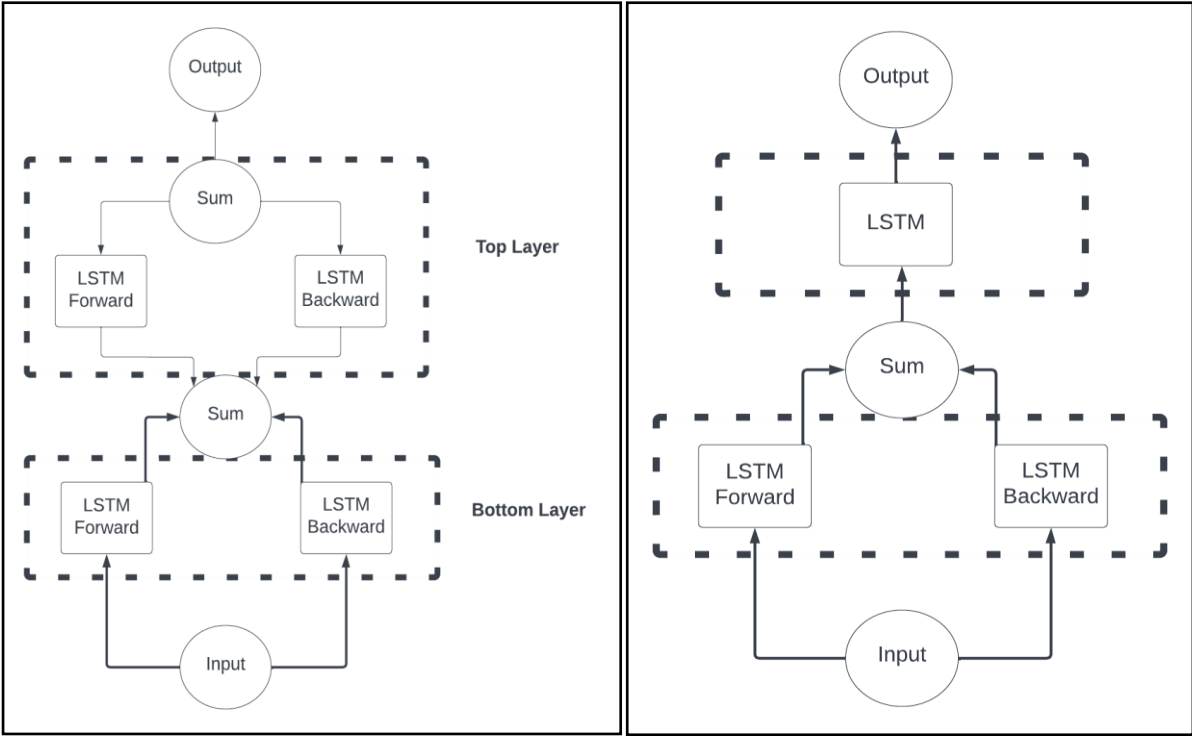


Figure 5. Bidirectional LSTM Encoder and Decoder

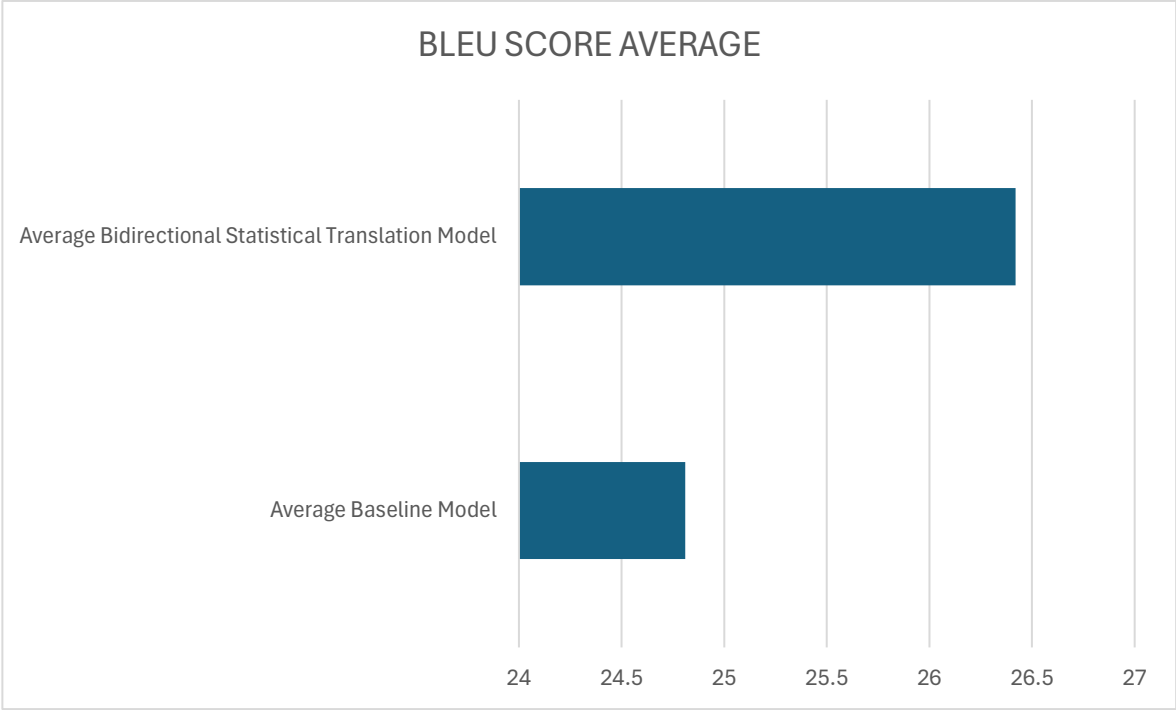


Figure 6. Bleu Score Average

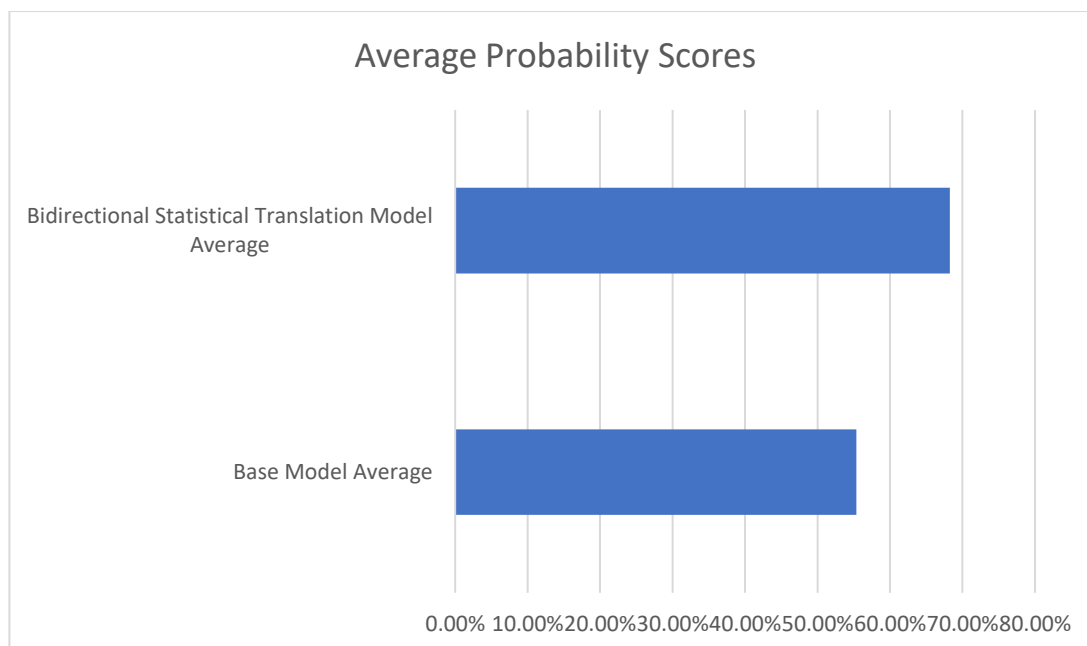


Figure 7. Average Probability Score

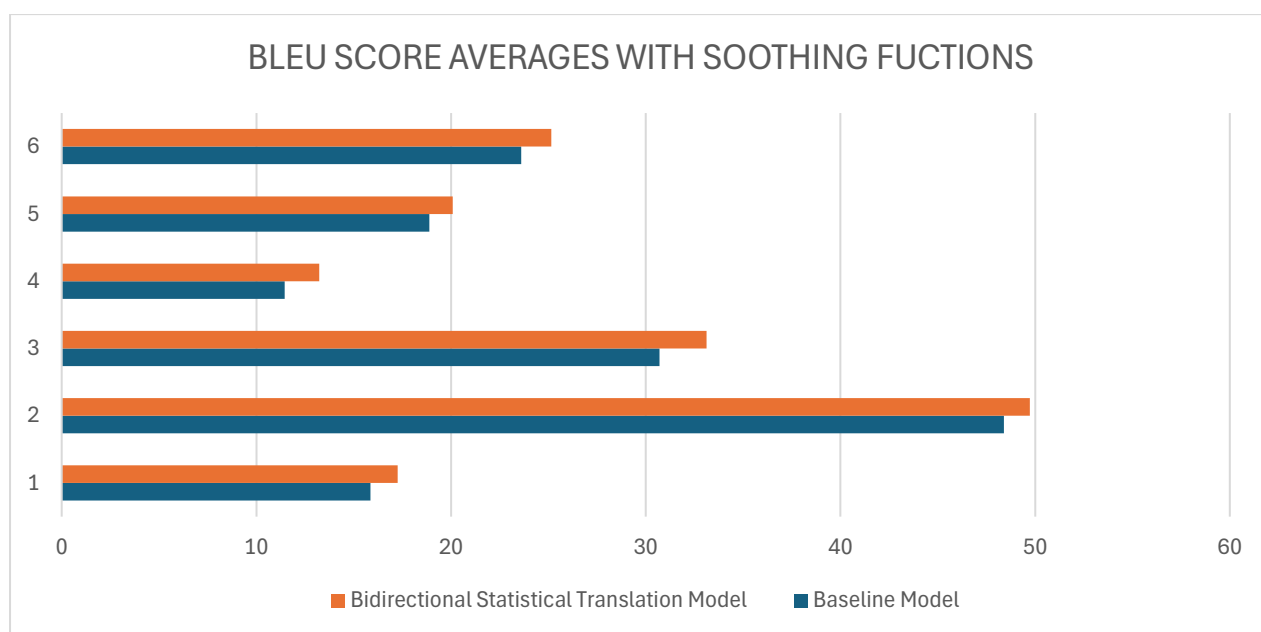


Figure 8. BLUE Score Average with Soothing Functions

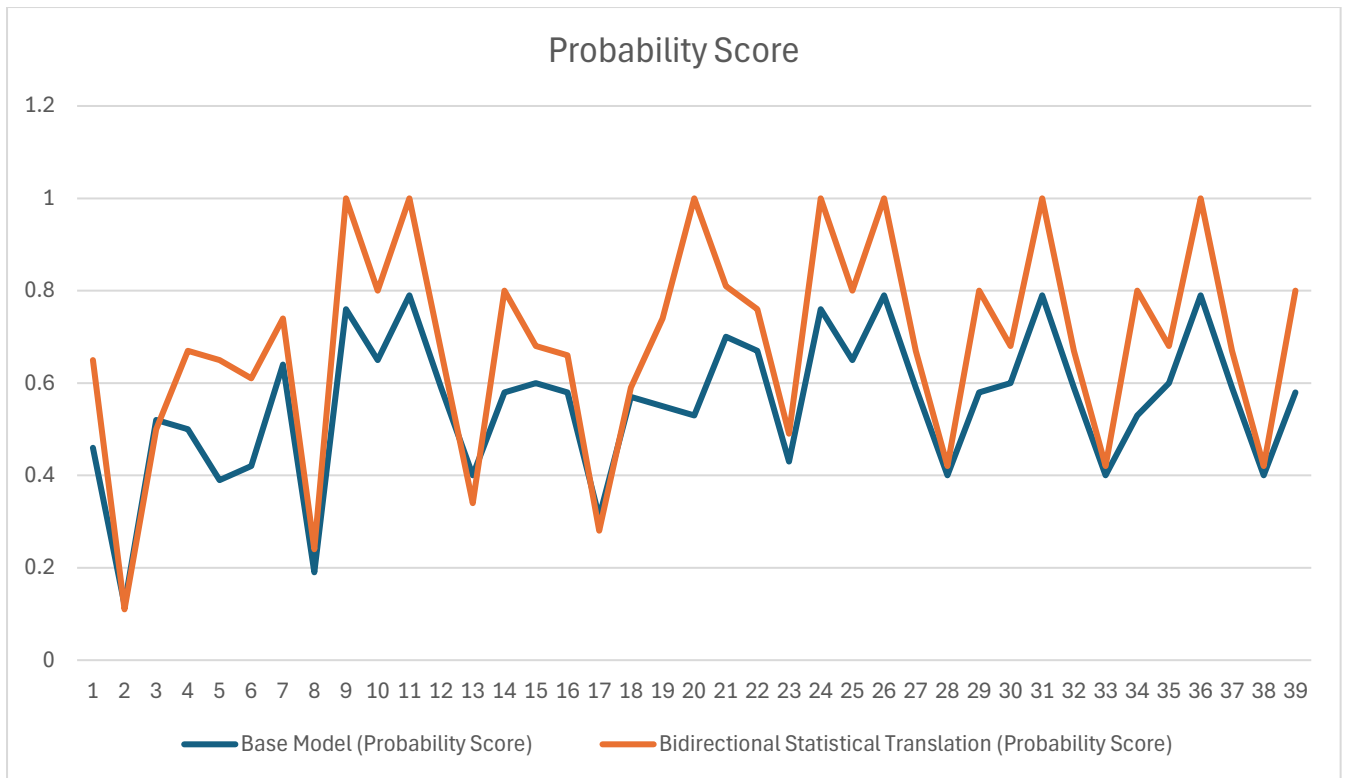


Figure 9. Probability Score

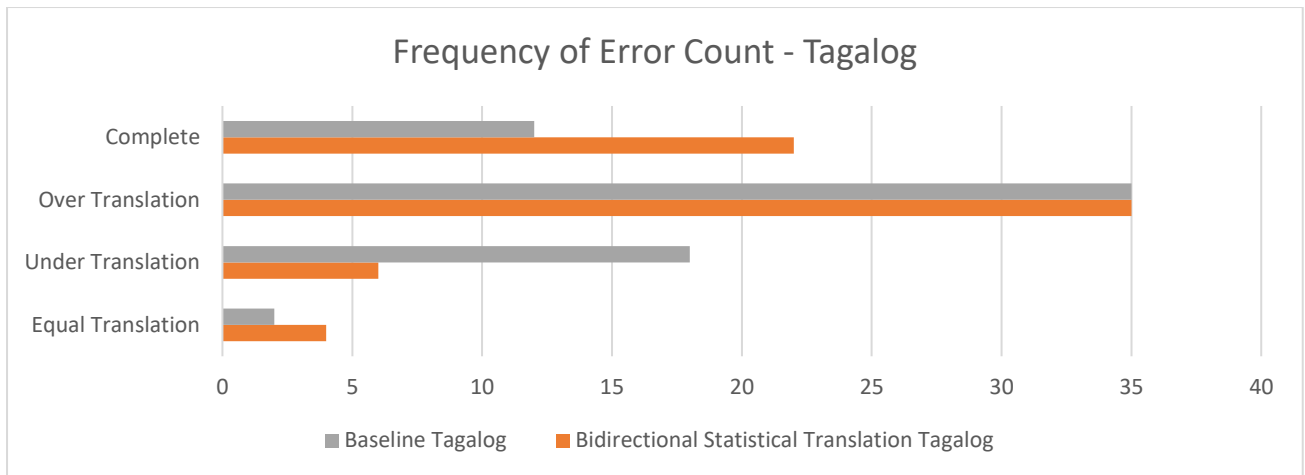


Figure 10. Frequency of Error Count - Tagalog

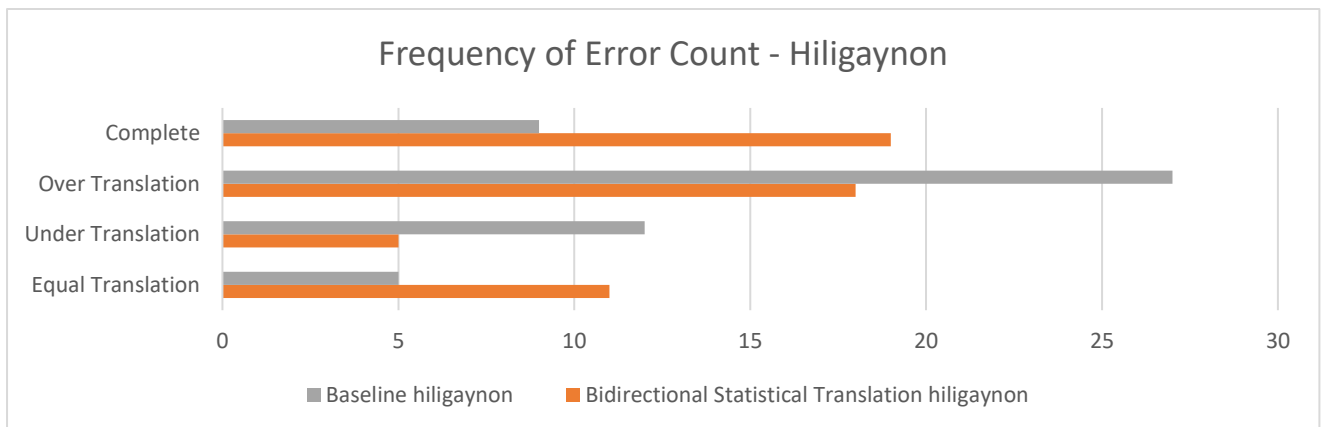


Figure 11. Frequency of Error Count - Hiligaynon

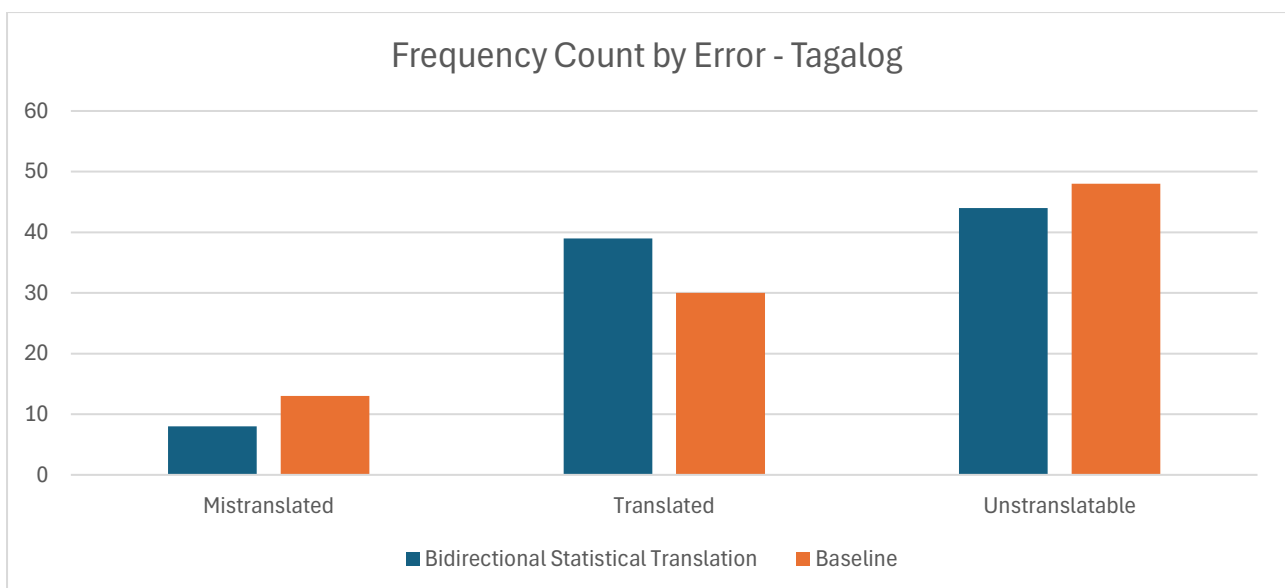


Figure 12. Frequency Count by Error - Tagalog

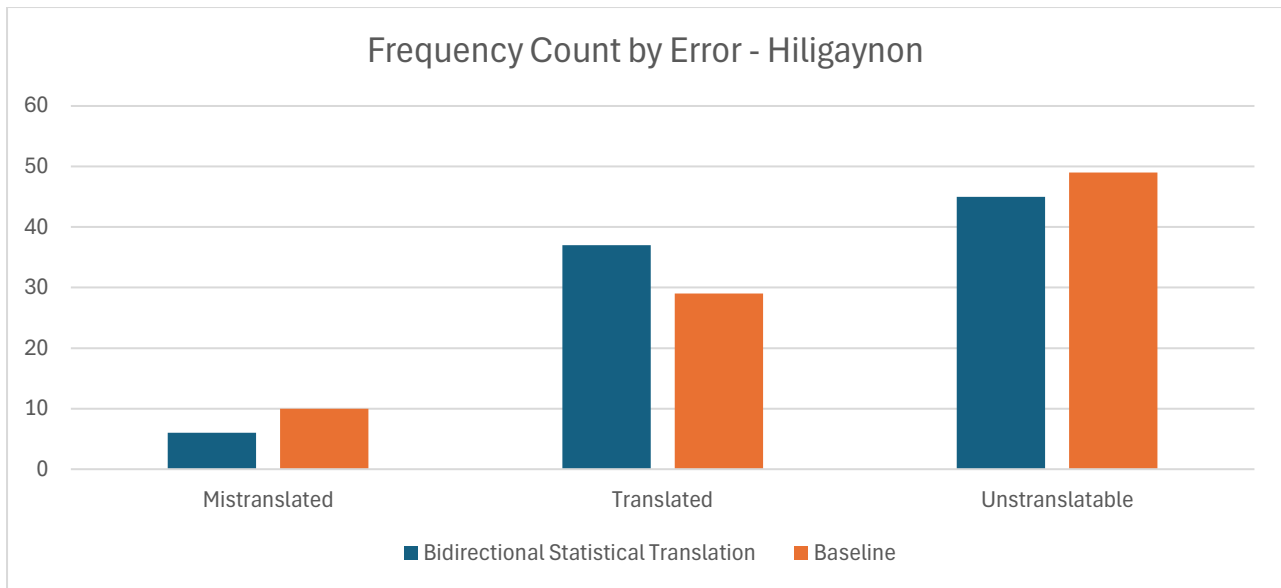


Figure 13. Frequency Count by Error - Hiligaynon

Base Model	Bidirectional Statistical Translation Model
15.86	17.25
48.39	49.73
30.71	33.12
11.45	13.23
18.88	20.08
23.61	25.15

Table 1. BLUE Score Average with Soothing Functions

Frequency of Error Count - Tagalog	Bidirectional Statistical Translation Tagalog	Baseline Tagalog
Equal Translation	4	2
Under Translation	6	18
Over Translation	35	35
Complete	22	12

Table 2. Frequency of Error Count - Tagalog

Frequency of Error Count - Hiligaynon	Bidirectional Statistical Translation Hiligaynon	Baseline Hiligaynon
Equal Translation	11	5
Under Translation	5	12
Over Translation	18	27
Complete	19	9

Table 3. Frequency of Error Count - Hiligaynon

Frequency Count by Error - Tagalog	Bidirectional Statistical Translation	Baseline
Mistranslated	8	13
Translated	39	30
Untranslatable	44	48

Table 4. Frequency Count by Error - Tagalog

Frequency Count by Error - Hiligaynon	Bidirectional Statistical Translation	Baseline
Mistranslated	6	10
Translated	37	29
Untranslatable	45	49

Table 5. Frequency Count by Error - Hiligaynon

APPENDIX H

Grammarly and Turnitin Results

Report: Thesis 2 - Finals

Thesis 2 - Finals

by Ramcis Vilchez

General metrics

36,420	5,367	344	21 min 28 sec	41 min 17 sec
characters	words	sentences	reading time	speaking time

Score



This text scores better than 96%
of all texts checked by Grammarly

Writing Issues

87	10	77
Issues left	Critical	Advanced

Plagiarism

This text hasn't been checked for plagiarism

A Deep Learning-Enhanced Statistical Machine Translation of Filipino Local Languages

ORIGINALITY REPORT

9%

SIMILARITY INDEX

2%

INTERNET SOURCES

3%

PUBLICATIONS

6%

STUDENT PAPERS

PRIMARY SOURCES

1

Submitted to Sunnyside High School

Student Paper

5%

2

"Natural Language Processing and Chinese Computing", Springer Science and Business Media LLC, 2018

Publication

1%

3

sylvester.bth.rwth-aachen.de

Internet Source

<1%

4

Vasco Alves, Jorge Ribeiro, Pedro Faria, Luis Romero. "Neural Machine Translation Approach in Automatic Translations between Portuguese Language and Portuguese Sign Language Glosses", 2022 17th Iberian Conference on Information Systems and Technologies (CISTI), 2022

Publication

<1%

5

staging-medinform.jmir.org

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6

Peter I. Udenze, Jiaqi Gong, Shohreh Soltani, Dawen Li. "A deep neural network with two-step decomposition technique for predicting ultra-short-term solar power and electrical load", Applied Energy, 2025

Publication

<1%

7

arxiv.org

Internet Source

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8

Submitted to Anna University

Student Paper

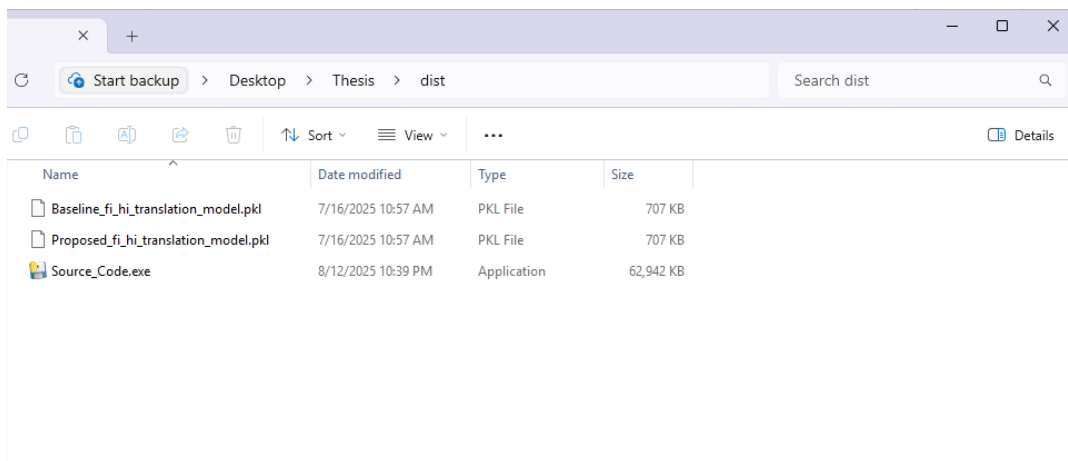
<1%

APPENDIX H

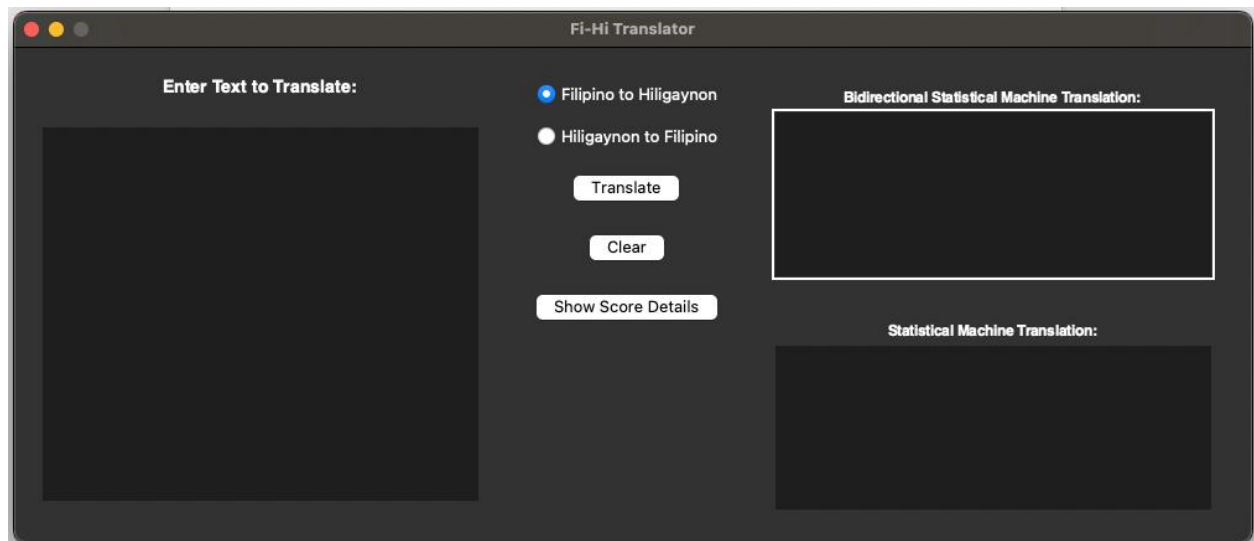
User Manual

USER MANUAL

To use the program, you just have to open the executable file Source_Code.exe located in the /dist folder. Do not move the file out of this folder. Do not delete the pkl files either.



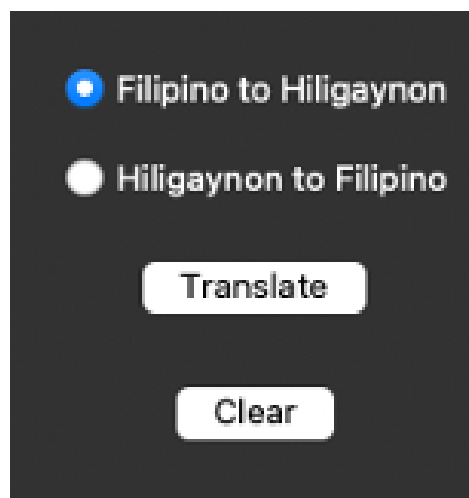
This is the User Interface:





A dark-themed application window with a title bar containing three colored buttons (red, yellow, grey). The window has a header labeled "Enter Text to Translate:". Below the header is a large text input area with a white border. The text "ikaw at ako" is entered at the top left of this area.

Enter the words on this text box



A dark-themed application window showing two radio button options. The first option is "Filipino to Hiligaynon" with a blue radio button that has a white dot in the center. The second option is "Hiligaynon to Filipino" with a grey radio button that is empty. Below the options are two white buttons with black text: "Translate" and "Clear".

Select the desired language to translate and then click the Translate button

Proposed Translation:

ikaw kag ako

Baseline Translation:

ikaw kag ako

The translation will be displayed on this area of the program. It can be copied and paste to a text file

☒ Filipino to Hiligaynon

☐ Hiligaynon to Filipino

Translate

Clear

Proposed Translation:

Baseline Translation:

Click the Clear button to clear out the translation text boxes



Click the Show Score details to display the scores in terms of Quality, Probability and BLEU.